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A GEOMETRIC DIAGNOSIS OF CROSS-CORPUS FAILURE IN SPEECH
EMOTION RECOGNITION

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Introduction

Motivation

Speech emotion recognition is an automated task that seeks to infer an emotional label for a spoken utterance based on its acoustic signal. Current approaches typically rely on self-supervised models such as wav2vec 2.0 [1], which are first pretrained on large collections of unlabelled audio and then fine-tuned on emotion-labelled datasets. While these systems achieve high accuracy on the dataset used for fine-tuning, they usually perform poorly on other datasets, even when the set of emotion categories is the same.

This cross-corpus failure has been documented [2, 3]. The literature has so far treated this as a problem of domain shift to be mitigated through data augmentation, multi-corpus training, or adversarial adaptation. This work instead is devoted to a prior question: *where exactly does the failure originate in the trained model, and what is its geometric structure in the learned representation space?*

Self-supervised encoders such as wav2vec 2.0 are often treated as black boxes, which is used to justify skipping diagnosis and retraining end to end. This thesis takes the opposite view. By probing the internal geometry of a wav2vec 2.0 model fine-tuned for emotion recognition, it isolates where cross-dataset failure occurs, characterizes it geometrically, and supports the diagnosis with a minimal intervention that leaves the encoder unchanged.

Goals

The thesis pursues three interconnected goals. The first is *diagnostic*: to determine whether the cross-dataset failure of a fine-tuned wav2vec 2.0 model originates in the encoder, implying that the representation cannot reflect emotions outside its training distribution, or in the classification head, whose decision boundaries are tied to the training data and may be misaligned with the target representation.

The second is *constructive*: to translate this finding into a practical adaptation strategy. If the encoder representation remains largely intact and only the boundaries are misaligned, a small

classification head trained on target-domain labels, with the encoder frozen, should suffice to recover the lost performance.

The third is *evaluative*: to test how the resulting system transfers across languages. The frozen wav2vec 2.0 backbone is fine-tuned on English and the head is trained on English and Chinese; the model is then evaluated, without further adaptation, on German, French, and Greek corpora, languages that received no emotional supervision at any stage, though they belong to the multilingual data on which the backbone was pretrained.

Structure of the thesis

The thesis is organized into six chapters that follow the order of the analysis described above.

Chapter 1 introduces the problem of speech emotion recognition and reviews the state of the art, covering emotion representation models, classical probabilistic methods, and modern deep learning architectures for the task.

Chapter 2 presents the fundamentals of audio signal processing and an exploratory analysis of conventional acoustic features on the considered corpora.

Chapter 3 presents the wav2vec 2.0 model, its self-supervised pretraining objective, and the architectural components whose outputs serve as the embedding representation for the rest of the thesis.

Chapter 4 constitutes the diagnostic core of the thesis. It characterizes the cross-dataset transfer failure of a baseline model on ESD and evaluates two hypothesized mechanisms: (H1) the encoder fails representing emotion for inputs that fall outside its training distribution, and (H2) emotional information remains encoded, but the decision boundaries learned on RAVDESS are misaligned with the representation space induced on ESD.

Both hypotheses are investigated via geometric analyses of encoder embeddings. Linear probing falsifies the strong form of H1 by demonstrating that emotional information remains linearly decodable. Principal-angle analysis further reveals that the emotion subspace is approximately orthogonal to the language, gender, and speaker subspaces, while ablation experiments confirm that the residual overlap among these subspaces does not convey emotional content. Finally, a Procrustes alignment analysis implicates misalignment as the dominant failure mode, indicating that a single orthogonal rotation accounts for a substantial portion of the directional discrepancy between the RAVDESS-trained decision boundaries and the ESD embedding space.

Chapter 5 turns this diagnosis into an adaptation strategy: a compact classification head is trained on the frozen wav2vec 2.0 backbone using English and Chinese ESD labels. It then tests the cross-linguistic transfer established earlier: with the encoder frozen, the head is evaluated zero-shot on three emotional speech corpora, EmoDB (German), CaFE (French), and AESDD (Greek), none seen during backbone fine-tuning or head training.

Chapter 6 draws the conclusions of the thesis. It retraces the diagnostic, constructive, and evaluative line of reasoning, and relates the cross-corpus results to an acoustic-level analysis. It then outlines future work, centered on connecting the macroscopic geometry of the encoder representation to the prosodic and spectral properties of the speech signal.

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Chapter 1

State of the art in speech emotion recognition

Speech Emotion Recognition (SER) is an area of Affective Computing that seeks to enable machines to detect human affect states using formal methods. The term *Affective Computing* [4] was formally defined by Rosalind Picard, who defined the properties that intelligent agents should have to understand and react to human emotional states.

1.1 Models of emotion representation

In the domain of affective computing, one central challenge is how emotional classes can be encoded in a human-readable format. Two primary approaches to representing emotions have emerged [4]. The first relies on categorical labels that represent basic emotions, while the second uses continuous dimensions to characterize emotions as the outcomes of continuous physiological processes.

The research of Paul Ekman [5] constitutes a reference point in the scientific study of emotion and has provided a foundation for numerous computational modeling approaches. Through empirical analysis of facial expressions, Ekman identified several distinct facial expression patterns associated with emotional communication. His cross-cultural studies demonstrated that specific emotions, such as anger, disgust, fear, happiness, sadness, and surprise, can be recognized in different cultural contexts. In 1992, this work was formally consolidated into a theoretical framework that defines these emotional states as *basic emotions*.

A substantial number of datasets and computational models have directly adopted this taxonomy for classification tasks. Nevertheless, despite their influence, these discrete categories can be restrictive, especially in contexts where emotional states do not conform to fixed labels or where nuances and gradations of affective experience are of central importance.

In contrast to categorical models, dimensional models, which began to emerge around 1980 [6, 7], represent emotions along continuous scales. A widely used dimensional framework posits three core dimensions on which emotions can be located:

1. *Valence*, the extent to which an emotion is experienced as positive or negative;
2. *Arousal*, the level of activation or excitement;
3. *Dominance*, the extent to which an emotional state involves feelings of power versus submission.

For example, sadness is typically a low-arousal, negative emotion with a low degree of dominance, while anger is negative and high in arousal, with a high degree of dominance [8].

Although these representational schemes are useful, several challenges hinder robust emotion recognition. Firstly, there is no universally accepted definition of emotion [9]. Culture, personal experience, and psychological traits shape the way people express and communicate emotions. This variability is problematic for supervised learning, which is based on a reliable ground truth. When labels are ambiguous, that uncertainty is learned and reproduced by models.

This limitation has recently driven a shift toward methods that learn emotion representations from unlabeled data using self-supervised learning [10, 11]. Instead of relying on human annotation, unsupervised and self-supervised learning models infer structure directly from the audio signal, producing representations that capture key aspects of emotional expression. A prominent example is the self-supervised model Wav2Vec 2.0, which learns speech representations from unlabeled audio and can then be fine-tuned for SER using only a small amount of labeled data.

1.2 Probabilistic modeling: Hidden Markov Models

A Hidden Markov Model [12] offers a probabilistic framework to characterize the link between latent emotional states and the acoustic characteristic that transmits them indirectly. More precisely, a Hidden Markov Model is defined by the following elements:

Let N be the number of emotional states represented in the model. These states constitute the set:

$$\mathcal{Q} = \{q_1, q_2, \dots, q_N\}.$$

The model operates on an observation sequence:

$$\mathcal{O} = (O_1, O_2, \dots, O_T),$$

where T indicates the total number of time frames in the speech signal and each observation is an acoustic features vector.

An HMM is characterized by three probability distributions:

1. **State transition probabilities**, which model how emotional states evolve over time within a single utterance:

$$A = \{a_{ij}\}, \quad a_{ij} = \Pr(q_{t+1} = q_j \mid q_t = q_i),$$

where q_t denotes the hidden emotional state at time t .

2. **Observation emission probabilities**, which associate each emotional state with its observable acoustic realizations. Let

$$V = \{v_1, v_2, \dots, v_M\}$$

denote the set of possible observation symbols (e.g., quantized acoustic feature vectors), where the index $k \in \{1, \dots, M\}$ identifies a specific observation. The emission probabilities are defined as:

$$B = \{b_j(k)\}, \quad b_j(k) = \Pr(o_t = v_k \mid q_t = q_j),$$

representing the probability of observing the symbol v_k at time t given that the system is in an emotional state q_j .

3. **Initial state probabilities**, which describe the distribution over emotional states at the beginning of an utterance:

$$\pi = \{\pi_i\}, \quad \pi_i = \Pr(q_1 = q_i).$$

The complete HMM is specified concisely by the parameter set:

$$\lambda = (A, B, \pi).$$

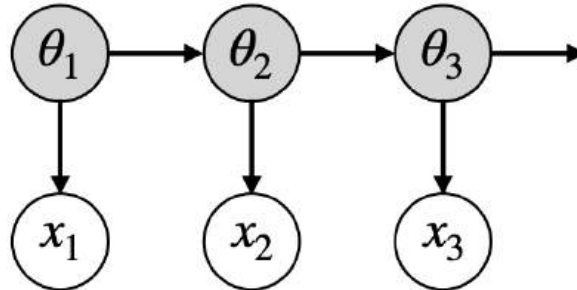


Figure 1: The hidden states are represented by gray nodes, while the observed data are shown as white nodes. Figure adapted from: OSL Dynamics documentation, *Hidden Markov Model (HMM)*, <https://osl-dynamics.readthedocs.io/en/latest/models/hmm.html>

1.2.1 Latent emotional states and observability

Before deep learning became widespread, Hidden Markov Models [4, 12] were central to speech emotion recognition. They treated emotion as a latent variable inferred from observable acoustic features and provided a probabilistic framework for modeling how emotional states evolve over time.

Hidden Markov Models describe emotion recognition using two stochastic processes:

1. *The hidden state sequence* is a chain of emotional states that are not directly observable;
2. *The observable sequence* is a representation of acoustic properties whose values depend probabilistically on the hidden states.

This methodology is applied to emotion recognition under the assumption that the acoustic manifestations of an inner affective state can be measured, even though the state itself is not directly observable. Given a sequence of acoustic observations, the model infers the most probable underlying sequence of latent emotional states that could have generated these observations. In this way, it models the temporal evolution of emotions, treating the acoustic manifestations of speech as a probabilistic manifestation of the underlying emotional state.

1.3 Deep representation learning for speech emotion recognition

Traditional SER systems are largely dependent on manually engineered features, but the rise of deep learning introduced a major shift: a variety of techniques [13] that can automatically learn optimal representations directly from raw acoustic input have emerged. Three principal architectural families have become foundational to deep representation learning for speech emotion recognition [14]: convolutional neural networks (CNN), recurrent neural networks (RNN) such as long short-term memory models (LSTM) and hybrid CNN–RNN architectures that integrate both paradigms [15].

More recently, Transformer-based architectures have emerged as a competitive alternative. Transformers model temporal and contextual dependencies using the self-attention mechanism that can capture long-range and global dependencies within an utterance, without iterative step-by-step processing, thus enabling fully parallel computation over the input sequence.

1.3.1 Convolutional architectures for spectrogram analysis

Convolutional Neural Networks (CNNs) [16] are designed to extract informative patterns that are distributed across an input signal. Depending on the input representation, CNNs can localize patterns along the temporal dimension, as in one-dimensional time-series signals, or jointly

in the time–frequency domain when operating on spectrogram-based representations. This capability is enabled by the convolution operation, which allows for local pattern detection while preserving the underlying structure of the input.

A convolutional layer consists of a set of learnable kernels (or filters) that act as pattern detectors. Given an input signal x and a filter w , the convolution operation computes a feature map y as:

$$y(t) = (x * w)(t) = \sum_{\tau} x(\tau) w(t - \tau),$$

where $*$ denotes convolution. In practice, this corresponds to computing a weighted sum over local regions of the input as the filter is translated across all valid positions.

Each filter is applied across the entire input domain, producing a distinct feature map whose activations indicate the presence and location of the corresponding pattern. If a convolutional layer contains K filters, it produces a set of feature maps:

$$\mathbf{Y} = \{y^{(1)}, y^{(2)}, \dots, y^{(K)}\},$$

which together form a transformed representation that is propagated to subsequent layers of the network. During training, the filter parameters are learned by minimizing a task-dependent loss function $\mathcal{L}$ using gradient-based optimization:

$$w \leftarrow w - \eta \nabla_w \mathcal{L},$$

where η is the learning rate.

CNNs exhibit a hierarchical structure as each processing stage constructs increasingly abstract representations. Early convolutional layers typically capture simple and local regularities. As additional layers are stacked, the filters operate on previously extracted feature maps, enabling the composition of more complex, higher-level structures. Two key mechanisms influence the evolution of these hierarchical representations:

- **Padding** is introduced to preserve the spatial or temporal resolution of the input. Without padding, convolutional operations near the boundaries would have access to less contextual information and successive convolutions would progressively shrink the representation. Formally, padding extends the input x to $\tilde{x}$ by adding zeros (or other values) at its borders before convolution.
- **Pooling** reduces the spatial or temporal resolution of the feature maps while retaining the most salient information. For example, max pooling computes:

$$y_{\text{pool}}(t) = \max_{\tau \in \mathcal{R}(t)} y(\tau),$$

where $\mathcal{R}(t)$ denotes a local neighborhood. This operation introduces a controlled compression on the representation.

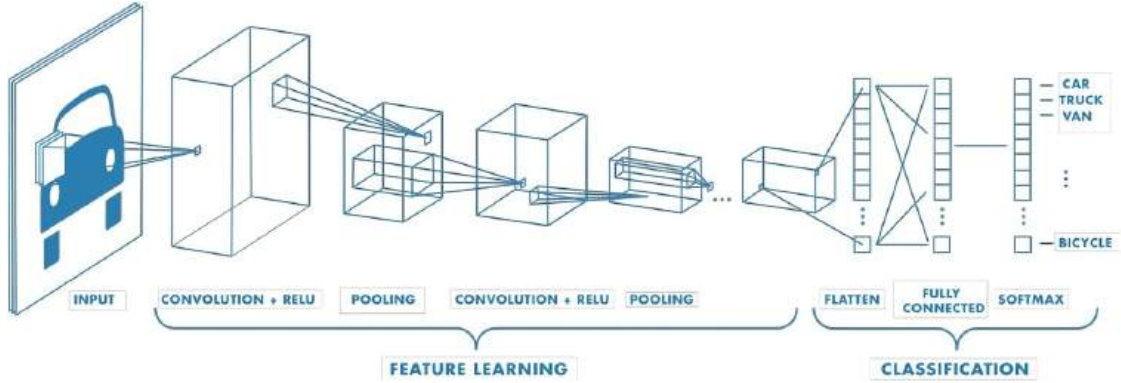


Figure 2: Architecture of convolutional neural networks. Figure source: MathWorks, *What Is a Convolutional Neural Network?*, available at <https://www.mathworks.com/discovery/convolutional-neural-network.html>

For prediction, a common strategy consists of flattening the final set of feature maps into a single vector, which is then passed onto one or more fully connected layers. These dense layers are responsible for associating specific configurations of extracted features with the target output.

The structure of the output layer depends on the nature of the task. In binary classification, the network typically produces a single scalar z , which is assigned to a probability using the sigmoid function:

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

In multi-class classification, the network produces a vector of class scores $\mathbf{z}$, which is transformed into a probability distribution via the softmax function:

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}.$$

1.3.2 Recurrent architectures for temporal modeling

Recurrent Neural Networks (RNNs) [17] are a class of deep learning architectures specifically designed to model sequential data by explicitly incorporating temporal ordering into the computation. This property is particularly important in audio processing, where information evolves over time and the interpretation of a given acoustic frame is strongly influenced by its surrounding temporal context.

Unlike conventional feedforward neural networks, which require inputs of fixed dimensionality, RNNs naturally accommodate input sequences of variable length. This characteristic is aligned well with speech and audio signals, where utterances may differ in duration.

A RNN processes an input sequence $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\}$ one element at a time while maintaining a

hidden state that acts as a form of dynamic memory. At each time step t , the hidden state $\mathbf{h}_t$ is updated by combining the current input with the hidden state from the previous time step:

$$\mathbf{h}_t = f(W_x \mathbf{x}_t + W_h \mathbf{h}_{t-1} + \mathbf{b}),$$

where $f(\cdot)$ is a nonlinear activation function, and W_x , W_h , and $\mathbf{b}$ are learnable parameters.

The updated hidden state $\mathbf{h}_t$ is then propagated forward in time, allowing information from earlier inputs to influence subsequent computations. The same recurrent unit and parameter set are reused at every time step, implying that:

$$(W_x, W_h, \mathbf{b}) \text{ are shared for all } t.$$

This parameter sharing enforces a consistent temporal processing mechanism and enables the model to learn temporal dependencies directly from the data during training.

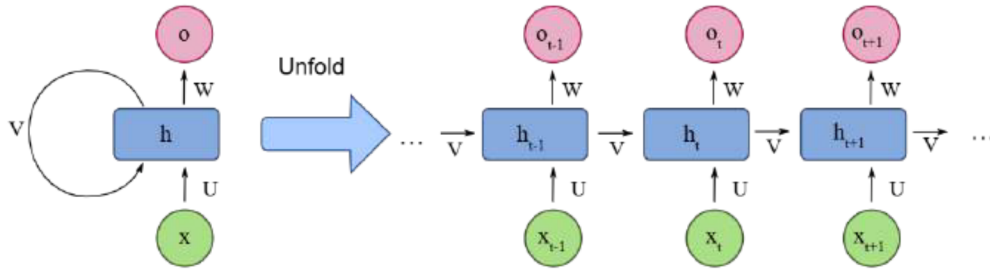


Figure 3: Folded and unfolded representations of a recurrent neural network. Figure source: *Recurrent neural network*, available at https://it.wikipedia.org/wiki/Rete_neurale_ricorrente.

Since the hidden state at each time step depends on all preceding states, the model parameters affect the output throughout the sequence. Training therefore requires propagating gradients backward through time across multiple sequential computations, a procedure known as back-propagation through time (BPTT). Formally, the loss $\mathcal{L}$ accumulated over the sequence depends on all hidden states $\{\mathbf{h}_t\}_{t=1}^T$, and the gradients must be computed through repeated applications of the state transition function.

This training process can be computationally intensive and is susceptible to vanishing or exploding gradients, especially for long input sequences. These challenges have motivated the development of gated recurrent architectures, such as Long Short-Term Memory (LSTM) networks. More recently, these limitations have also contributed to a shift toward attention-based and transformer architectures.

1.3.3 Self-attention and transformer architectures

The Transformer architecture [18] represents a fundamental shift in sequential modeling. Unlike recurrent or convolutional approaches, Transformers rely on attention mechanisms to model dependencies within a sequence. This design adapts well to audio processing tasks, where long-range temporal dependencies play a critical role, and efficient parallel computation is desirable.

Self-attention Self-attention enables the model to compute contextualized representations by explicitly measuring the relevance between pairs of sequence elements. For each time step, the representation is updated by selectively aggregating information from all other positions in the sequence, weighted by their estimated importance.

Directly computing relevance in the original embedding space would entangle all semantic components, including those irrelevant to the current context. To address this, each input embedding is projected into three distinct linear spaces, each serving a different functional role:

- A **query** vector Q , representing the information that a token seeks;
- A **key** vector K , representing the information that a token offers;
- A **value** vector V , containing the information that will be propagated if the token is relevant.

This separation allows the model to independently learn how relevance is computed and what information is exchanged once relevance has been established.

Formally, given an input embedding matrix $E \in \mathbb{R}^{T' \times d}$, linear projections are applied:

$$Q = EW_Q, \quad K = EW_K, \quad V = EW_V,$$

where $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ are learned parameter matrices. Relevance between tokens i and j is computed using dot products:

$$\text{score}(i, j) = Q_i K_j^\top.$$

To maintain numerical stability, the scores are scaled by the dimensionality of the key space:

$$\tilde{S} = \frac{QK^\top}{\sqrt{d_k}}.$$

A softmax function is applied row-wise to obtain normalized attention weights:

$$A = \text{softmax}(\tilde{S}),$$

which form a probability distribution over the sequence. The final contextualized representations are computed as a weighted sum of the value vectors:

$$\text{Attention}(Q, K, V) = AV.$$

Through this mechanism, each token selectively integrates information from the entire sequence based on learned relevance patterns.

Multi-head attention A single self-attention operation can model only one type of relational structure at a time. However, audio and language signals exhibit multiple simultaneous dependencies, such as local continuity, long-range temporal structure, and task-specific correlations.

Multi-head attention addresses this limitation by computing several self-attention operations in parallel. Each attention head applies its own set of projection matrices, producing independent queries, keys, and values:

$$\text{head}_h = \text{Attention}(EW_Q^{(h)}, EW_K^{(h)}, EW_V^{(h)}).$$

The outputs of all heads are concatenated and linearly transformed:

$$\text{MultiHead}(E) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W_O,$$

allowing the model to jointly attend to information from multiple representational subspaces.

Positional encoding Because Transformers process all sequence elements in parallel, they are inherently permutation-invariant and do not encode temporal order. However, temporal structure is essential in audio signals, where phonetic transitions, rhythm, and prosodic patterns must be considered over time.

To inject order information, positional encodings are added to the input embeddings before attention is applied:

$$X' = X + P,$$

where P is a position-dependent encoding.

Effective positional encoding must be bounded in magnitude so that positional information does not dominate acoustic content or destabilize training and must be uniquely represented to avoid ambiguity.

A Transformer is typically organized as an encoder–decoder architecture.

Encoder The encoder processes the entire input sequence simultaneously and produces a sequence of contextualized representations. After embedding the input tokens and adding positional encodings, the sequence is passed through a stack of identical layers. Each encoder

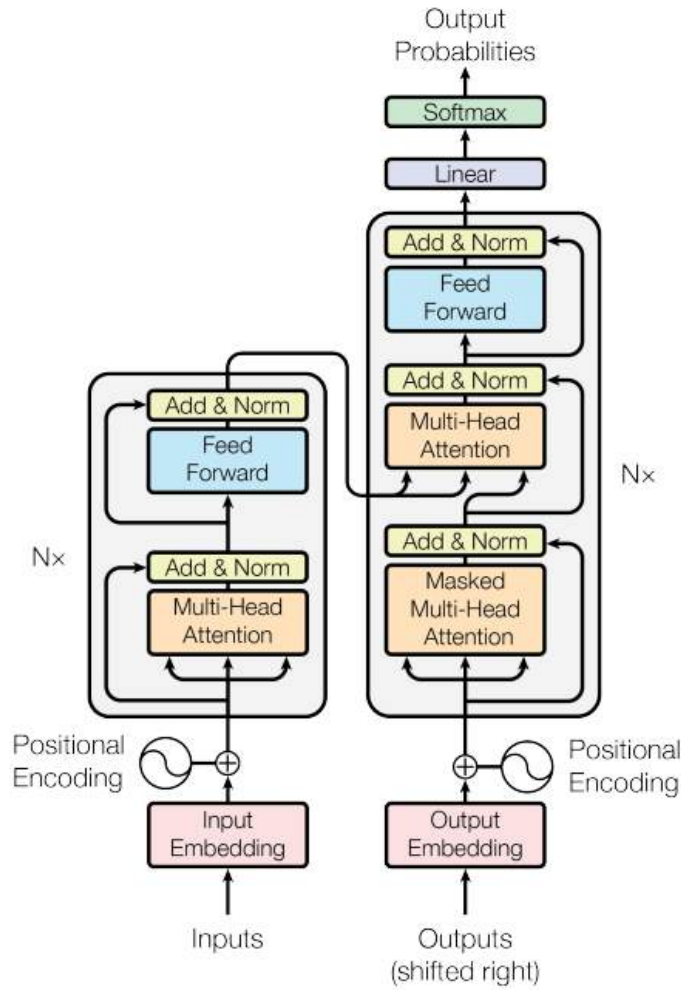


Figure 4: Architecture of a Transformer model with encoder and decoder components. Figure source: A. Vaswani et al., *Attention Is All You Need* [18].

layer consists of:

1. **Multi-head self-attention**, enabling global contextual interaction;
2. **Residual connections and layer normalization**, which stabilize training and facilitate gradient flow;
3. **Position-wise feed-forward networks**, which apply nonlinear transformations independently to each position;
4. **Residual connections and layer normalization**, ensuring stable signal propagation.

After multiple layers, the encoder produces a contextual memory, where each vector represents an input token enriched with information from the entire sequence.

Decoder The decoder generates the output sequence auto-regressively, producing one token at a time. Like the encoder, it operates on embedded tokens augmented with positional encodings, but introduces two additional mechanisms.

First, **masked multi-head self-attention** enforces causality by preventing each position from attending to future tokens. This is achieved by masking attention scores corresponding to positions $j > t$ before the softmax operation.

Second, the decoder employs **cross-attention**, where queries are generated from the decoder states, while keys and values are taken from the encoder outputs. This allows the decoder to selectively attend to relevant portions of the encoded input sequence during generation.

Stacking multiple decoder layers enables iterative refinement of the output representations. A final linear projection maps the decoder output to vocabulary-sized logits, which are converted into a probability distribution over possible next tokens using a softmax function.

Chapter 2

Acoustic representations of emotion in speech

The chapter presents an overview of fundamental audio processing techniques, with a focus on preprocessing and feature extraction tools that support exploratory data analysis and facilitate effective representation of audio signals for downstream machine learning tasks.

2.1 Time-frequency representations of the speech signal

Sampling is the process by which the recording system measures the analog sound wave a fixed number of times per second, known as the sampling rate, which determines the time resolution of the digital signal. After sampling, the measured signal is quantized, by mapping each sample amplitude to a value in a finite set of discrete values, allowing the representation of the continuous audio signal in a digital form.

Audio signals [19] can be analyzed in the time domain, where the horizontal axis denotes time and the vertical axis shows the amplitude of the air pressure. This representation illustrates how air pressure varies at each moment. However, on its own, this time-domain view has limited usefulness for most audio-processing tasks. To obtain more informative features, it is common to analyze it in terms of its constituent frequencies. The primary mathematical tool for this conversion is the Fourier transform, which decomposes a waveform into its frequency components. Its output consists of a series of complex numbers, each of which encodes:

- **Magnitude**, which indicates how much energy a specific frequency contributes to the signal;
- **Phase**, which determines the temporal alignment of that specific frequency.

When converting a waveform from the time domain to the frequency domain, the transform reveals how the signal's energy is allocated between different frequencies. However, a major limitation of the Fourier transform is that it discards information about when these frequencies occur in time. Real-world audio signals typically vary over time: a waveform may include relatively quiet segments with little activity, interspersed with segments exhibiting intense, rapid oscillations.

2.1.1 The spectrogram

To address the limitations of the standard Fourier transform, a representation that accounts for both the time domain and the frequency domain is employed: a spectrogram [20] is a visual display of an audio signal that shows how its frequency content evolves over time.

Spectrograms are produced using the Short-Time Fourier Transform (STFT), a construct that divides the audio into a sequence of short, consecutive time segments, called windows. A separate Fourier transform is then applied to each window, and the resulting spectrum from each segment becomes a vertical slice in the spectrogram.

The appearance of a spectrogram depends largely on the choice of window, which is a key determining parameter for what the spectrogram displays:

- Larger windows provide more accurate information about the frequency content, but tend to smooth rapid changes;
- Smaller windows offer better time resolution, but poorer frequency resolution.

Consequently, selecting a window size always involves balancing time resolution against frequency resolution.

Furthermore, once the signal is divided into segments, each segment is multiplied by a window function, a smooth curve that tapers the segment at its boundaries, to avoid the creation of discontinuities in the edges of the segment. One of the most commonly used window functions is the Hamming window. When a Hamming window is applied to an audio segment, it gradually reduces the amplitude near the edges, minimizing artifacts while maintaining the fidelity of the central region of the segment.

2.1.2 The mel spectrogram

The mel spectrogram is an alternative to the conventional spectrogram used in audio signal processing, particularly in tasks centered on human listening. It offers a spectral representation that better reflects the limitations and characteristics of the human auditory system.

The key distinction from a standard spectrogram lies in the frequency axis: a conventional spectrogram employs a linear frequency scale with uniformly spaced intervals, whereas human hearing responds to frequency in a non-linear manner, with higher sensitivity allocated to lower frequencies than to higher ones.

The mel scale itself is derived from empirical studies on human pitch perception. In these experiments, participants were asked to judge tones so that they appeared equally spaced in pitch, and a mathematical model was then constructed to fit these judgments, linking the physical pitch in Hertz to the perceived pitch in mels. The parameters used to transform frequencies into the mel scale are grounded in this empirical foundation, resulting in a nonlinear frequency scaling.

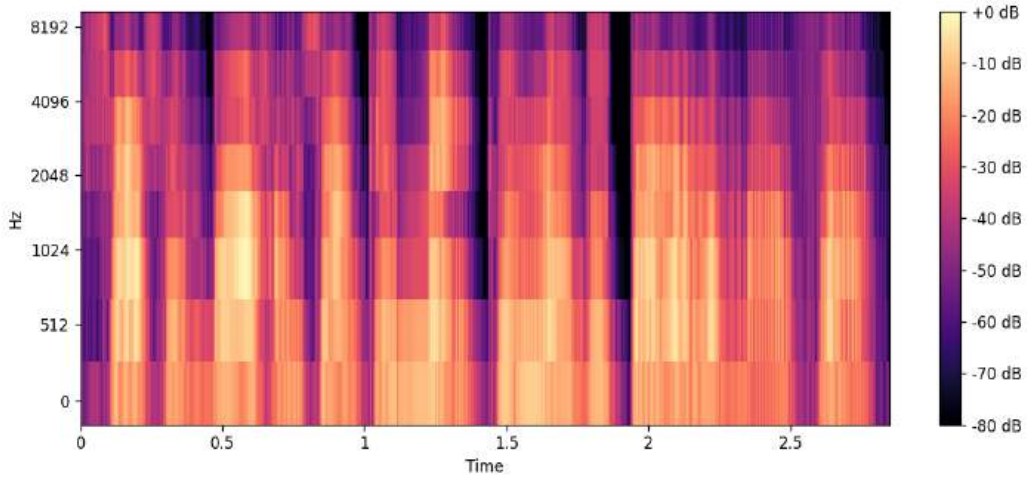


Figure 5: Mel spectrogram with a 4 ms analysis window ($n_{fft} = 64$ at 16 kHz). Time localization is excellent, every transient is sharply resolved along the time axis, while frequency resolution collapses: adjacent harmonics fuse into the same band.

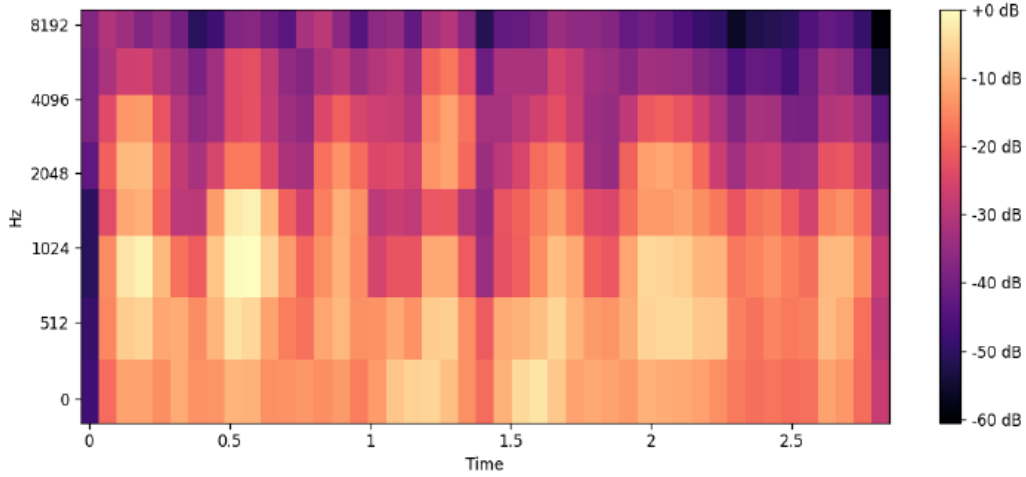


Figure 6: Mel spectrogram with a 256 ms analysis window ($n_{fft} = 4096$ at 16 kHz). Frequency resolution is excellent, harmonics appear as separated horizontal lines, while time localization collapses: each frame averages over a quarter second of audio, so transients smear across time and can no longer be precisely placed.

2.2 Exploratory analysis of acoustic features

Feature extraction consists of transforming audio signals into numerical representations that capture relevant acoustic properties. Then, EDA (Exploratory Data Analysis) is performed to investigate the statistical behavior of these features and their interdependencies, uncovering patterns linked to emotional categories. The following sections describe the extraction of acoustic features using the eGeMAPS feature set, followed by a data analysis of the resulting feature space.

2.2.1 The Geneva Minimalistic Acoustic Parameter Set

The Geneva Minimalistic Acoustic Parameter Set (*GeMAPS*) [21] was introduced as a standardized, interpretable, and reproducible baseline for voice analysis. GeMAPS specifies 18 low-level descriptors (LLDs), grouped into four principal categories:

- **Frequency-related features**, including the logarithm of the fundamental frequency and related measures of pitch;
- **Energy-related parameters**, which characterize vocal intensity through measures such as loudness;

- **Spectral balance features**, which quantify the distribution of acoustic energy across frequency bands, for example, the alpha ratio, contrasting energy in the 50–1000 Hz and 1–5 kHz ranges;
- **Temporal parameters**, which reflect speech rhythm and timing, including the durations of voiced and unvoiced segments and the syllable rate.

For each descriptor, a set of statistical functionals is computed, including the arithmetic mean, the standard deviation, and percentiles. This procedure yields a compact representation comprising 62 acoustic parameters.

The extended GeMAPS configuration (*eGeMAPS*) [21] was subsequently proposed to enhance recognition performance, incorporating seven additional LLDs, such as the first four Mel-Frequency Cepstral Coefficients, and the spectral flux, which quantifies frame-to-frame spectral variation. Together with their associated statistical functionals, these additional descriptors expand the parameter set to a total of 88 features.

2.2.2 The openSMILE toolkit

All acoustic features are extracted using the open-source openSMILE toolkit (*Speech and Music Interpretation by Large-space Extraction*) [22], which allows a reproducible implementation of the feature extraction pipeline.

OpenSMILE is implemented in C++ allowing for computational efficiency. Its architecture is component-based and can be flexibly configured via a single configuration file, facilitating repeatable experimental setups.

At the core of openSMILE is a central *Data Memory*, which functions as the primary communication interface among processing components. These components are organized into three main classes:

- *Data Sources*, which ingest input signals (e.g., pre-recorded audio files or live microphone streams);
- *Data Processors*, which compute acoustic features;
- *Data Sinks*, which export the resulting features to external targets such as CSV files or downstream machine learning modules.

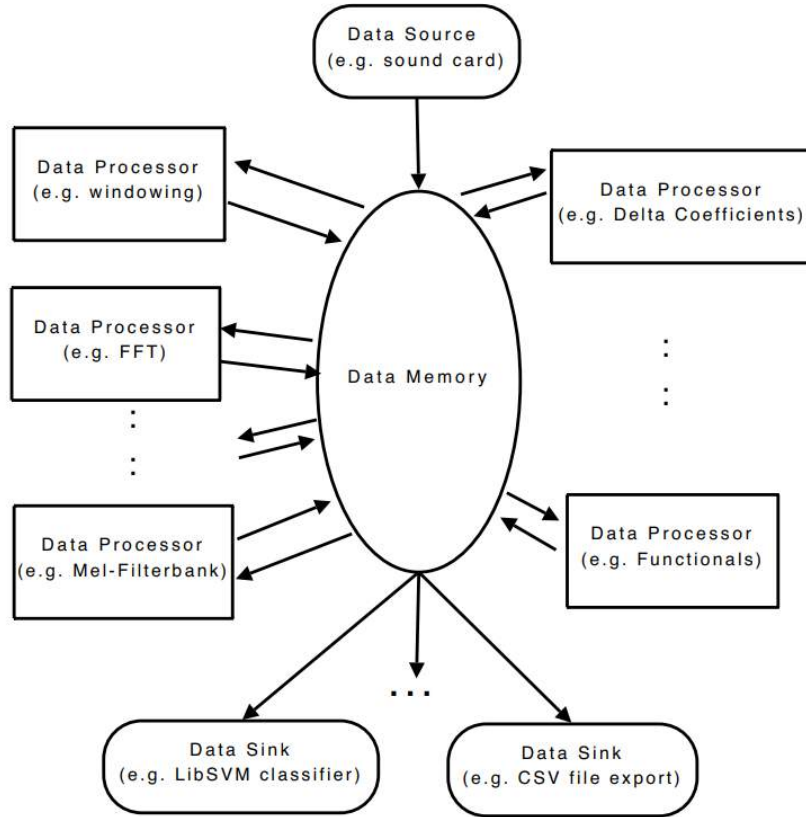


Figure 7: Diagram of openSMILE’s architecture. Figure source: F. Eyben, M. Wöllmer, and B. Schuller, *openSMILE – The Munich Versatile and Fast Open-Source Audio Feature Extractor* [22].

To further enhance computational performance, openSMILE supports multithreaded execution, allowing individual components to run in parallel on multi-core architectures. This design reduces processing time, particularly for large-scale corpora.

2.2.3 Emotional Speech Dataset (ESD)

The experiments described in this work are conducted using the Emotional Speech Dataset (ESD) [23], a public speech corpus designed for the investigation of emotional expression in spoken language. The dataset consists of recordings from 20 professional speakers, evenly divided between American English and Mandarin Chinese, with a balanced gender distribution. This balanced bilingual design facilitates the investigation of cross-lingual emotion recognition systems.

All speakers fall within the same age range (25–35 years), a design choice aimed at reducing confounding effects associated with age-dependent vocal characteristics. Linguistic variability is likewise controlled: all English speakers use North American English, whereas all Chinese

speakers use Standard Mandarin, thereby ensuring intralingual consistency within each language group.

Each speaker produced utterances corresponding to five discrete emotional categories: *Neutral*, *Happy*, *Angry*, *Sad*, and *Surprised*. In total, each speaker contributed 350 utterances per emotion, i.e. 1,750 utterances in total, yielding a balanced number of samples for every speaker–emotion combination. All recordings were made in a controlled indoor environment, with a signal-to-noise ratio greater than 20 dB and a sampling frequency of 16 kHz.

In this work, the dataset is exploited to analyze how different classes of acoustic features contribute to emotion discrimination at both the population level and the individual speaker level.

2.2.4 Feature standardization

Given the pronounced variability of speech signals between speakers, it is not expected that any single acoustic parameter can, in isolation, reliably differentiate between emotional states. Instead, emotional information is encoded in joint configurations of multiple acoustic features and is more informative when interpreted relative to the characteristic vocal behavior of an individual speaker, including habitual fundamental frequency range, vocal tract configuration, and speaking style.

For each speaker, features were standardized by subtracting the speaker-specific mean and dividing by the corresponding standard deviation, computed over all utterances produced by that speaker:

$$x' = \frac{x - \mu_s}{\sigma_s}$$

where μ_s and σ_s denote the mean and standard deviation, respectively, of the feature x for the speaker s .

This normalization procedure removes globally defined speaker-specific baselines while preserving relative intra-speaker variation within each individual’s expressive range.

2.2.5 Global variability of acoustic features across emotions

To obtain a characterization of the influence of emotional categories on acoustic realizations, a global multi-class analysis is conducted. Each acoustic feature extracted from the *eGeMAPS* set was assessed using ANOVA [24], with the emotion category treated as the independent factor.

For each feature, ANOVA compares how much the feature values differ between emotion categories against how much they vary within each category, through the F -statistic:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}, \quad (1)$$

where MS_{between} and MS_{within} denote the between- and within-group mean squares.

Intuitively, a large F means that the emotion-specific mean values are well separated relative to the natural spread of the feature inside each emotion, indicating that the feature genuinely reacts to the emotional category. A small F means that any apparent differences between emotions are dwarfed by the variability already present within a single emotion, and the feature is essentially uninformative.

Under the null hypothesis that emotion has no effect on the feature, the F -statistic follows a known distribution; the associated p -value quantifies the probability of observing an F value at least as large as the one measured purely due to random sampling variation. A small p -value (typically $p < 0.05$) therefore indicates that the observed dependency on emotion is unlikely to be a chance effect.

The p -value, however, only states whether an effect exists, not how large it is: with sufficiently many samples, even negligible differences become statistically significant. To quantify the magnitude of the effect we use eta squared η^2 , defined as the proportion of total variance in a given feature that is attributable to differences between emotion categories [25]:

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}, \quad (2)$$

where SS_{between} is the sum of squares explained by the emotion factor and SS_{total} is the total sum of squares. By construction $\eta^2 \in [0, 1]$ and admits a direct interpretation as the fraction of the feature’s variability that is explained by the emotional category.

Together, F and the associated p -value identify which acoustic features are significantly modulated by emotion, while η^2 ranks them by the strength of that modulation, providing a principled criterion to highlight the descriptors that carry the most emotion-discriminative information.

This global multi-class analysis, however, aggregates information across all emotions and all speakers, and therefore cannot reveal which specific emotional contrasts a feature actually discriminates, nor whether its discriminative behaviour is consistent across speakers. This motivates a more fine-grained analysis based on pairwise effect sizes and on speaker-specific contrasts, developed in the following section.

Table 1: Global separability of acoustic features across emotional classes, assessed via one-way ANOVA. Effect sizes are reported as $\eta^2 \times 100$ ($\eta^2 > 30\%$) and significance as p-values.

Feature	Effect Size (η^2)	p-value
F0 semitone from 27.5 Hz (80th percentile)	60.35	< 0.001
F0 semitone from 27.5 Hz (mean)	56.43	< 0.001
F0 semitone from 27.5 Hz (50th percentile)	50.08	< 0.001
MFCC1 V (mean)	44.71	< 0.001
Spectral flux V (mean)	37.79	< 0.001
F0 semitone from 27.5 Hz (20th percentile)	37.74	< 0.001
Spectral flux (mean)	37.45	< 0.001
Loudness mean falling slope	33.67	< 0.001
Alpha ratio V (mean)	33.39	< 0.001
Equivalent sound level (dBp)	30.97	< 0.001
F1 frequency (mean)	30.57	< 0.001
Loudness mean rising slope	30.52	< 0.001
Loudness (mean)	30.27	< 0.001

Since utterances are nested within speakers, these p-values should be read as indicative rather than as exact significance levels; the η^2 effect sizes and the speaker-level analysis below provide a more robust characterization

2.2.6 Pairwise separability of acoustic features across emotion pairs

This section offers a perspective on the exploratory analysis conducted at both the population and individual speaker levels. The results converge on two principal observations that recur throughout the analysis:

- No single acoustic feature exhibits fully robust discriminative power across emotional categories or across speakers;
- Analysis performed at the population level can obscure speaker-specific behaviors that diverge from aggregate patterns.

These findings suggest that global evaluations alone are insufficient to comprehensively characterize the variability of emotional speech; therefore, a complementary speaker-specific analysis is required to adequately capture complex expressive patterns.

Cohen's d

The degree of separation between each target emotion is quantified using Cohen's d [26], a measure defined as the difference between group means normalized by their pooled standard deviation.

$$d = \frac{\mu_A - \mu_B}{s_{\text{pooled}}}, \quad s_{\text{pooled}} = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}$$

where μ_A and μ_B denote the mean values of a given acoustic characteristic for the two emotional classes, s_A^2 and s_B^2 their corresponding variances and n_A , n_B the respective sample sizes.

Larger values of $|d|$ indicate stronger separation between the two distributions, while values approaching zero reflect distributional overlap and thus limited discriminative capacity.

For each emotion paired with neutral, the features extracted using *eGeMAPS* were ranked according to the absolute value of Cohen's d. To retain only the features exhibiting a substantial separation, a threshold of $|d| \geq 1$ was applied. This analysis characterizes how emotional states diverge from neutral speech and highlights the different acoustic mechanisms behind emotional expression.

Feature	Fear	Anger	Sadness	Happiness	Disgust
Speech rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grumbled chest tone
Pitch changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2: This table illustrates the variation of vocal features associated with different emotional affects relative to neutral speech (adapted from Murray and Arnott, 1993). [4, 27]

Exploratory Data Analysis at the population Level

In pairwise comparisons between neutral speech and the *categories of Surprise and Happy*, the acoustic feature *F0semitoneFrom27.5Hz_sma3nz_amean* demonstrates the strongest discriminative power among all evaluated acoustic dimensions, resulting in the clearest separation between emotionally expressive and neutral speech productions.

The speech segments labeled *Surprise* exhibit the greatest deviation along this dimension, with a Cohen's $|d| = 4.15$ relative to neutral speech. By comparison, *Happy* speech yields a value of $|d| = 3.225$ and *Angry* speech a value of $|d| = 2.276$.

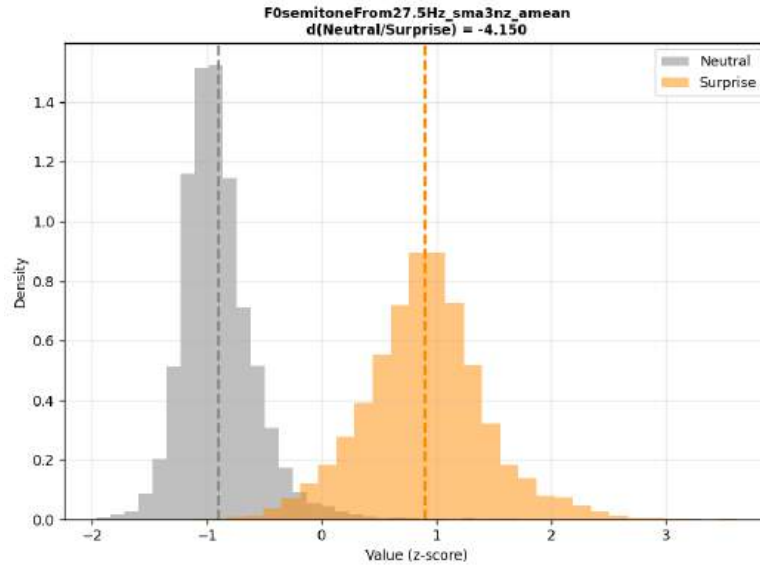


Figure 8: F_0 comparison between *Surprise* and *Neutral* speech

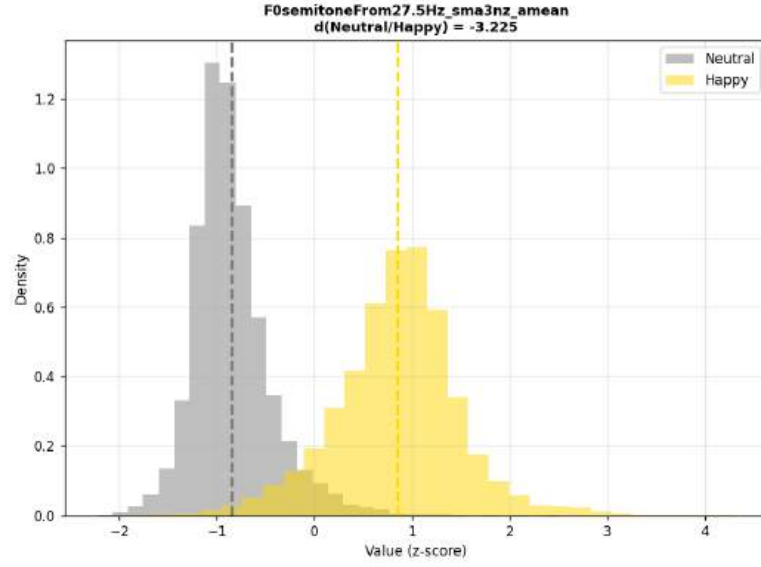


Figure 9: F_0 comparison between *Happy* and *Neutral* speech

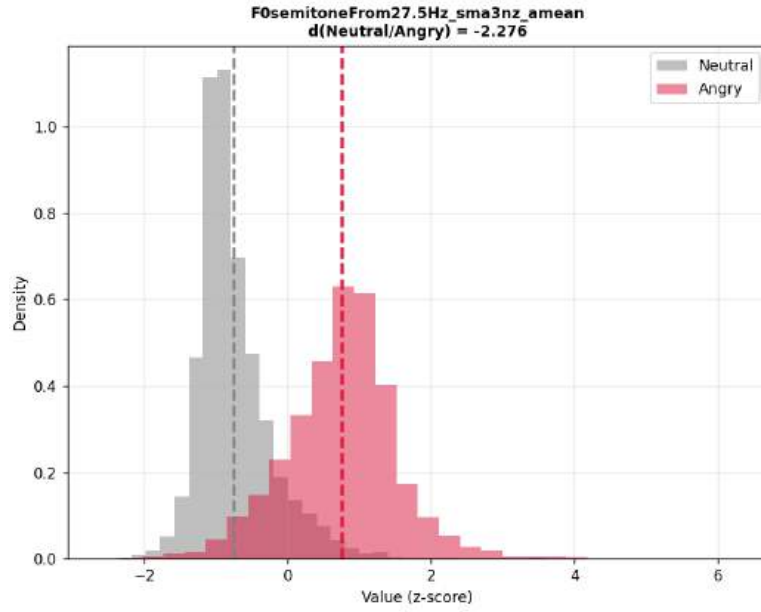


Figure 10: F_0 comparison between *Angry* and *Neutral* speech

Although the mean of the fundamental frequency F_0 exhibits discriminative power between neutral speech and the emotional categories *Surprise*, *Happy*, a closer examination of the corresponding feature distributions reveals only limited separability among these non-neutral emotional states.

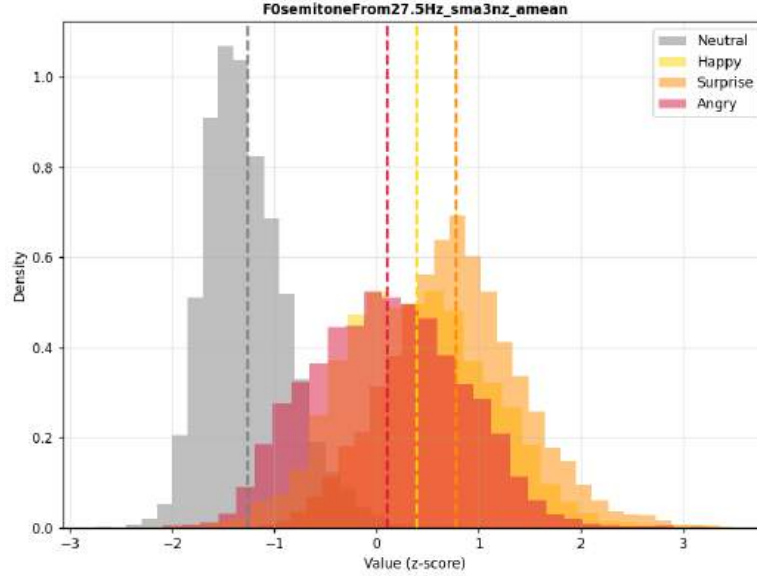


Figure 11: F_0 distribution of high-arousal affects

In general, the distributions exhibit substantial overlap and move toward higher values relative to neutral speech, indicating that the mean fundamental frequency provides limited discriminative power to differentiate among high-arousal states.

Table 3: Global separability of acoustic features across the *Happy*, *Surprise* and *Angry* emotional classes, assessed via one-way ANOVA and ordered by η^2 . Effect sizes are reported as $\eta^2 \times 100$ and statistical significance as p-values.

Feature	Effect Size (η^2)	p-value
F0 semitone from 27.5 Hz (80th percentile)	16.20	< 0.001
HNR dB ACF (mean)	14.20	< 0.001
MFCC3 (mean)	12.97	< 0.001
F0 semitone from 27.5 Hz (mean)	12.45	< 0.001
MFCC3 V (mean)	10.25	< 0.001
Loudness (std. dev. norm.)	9.06	< 0.001

This pattern is further supported by the one-way ANOVA performed exclusively on the three high-arousal categories *Happy*, *Surprise*, and *Angry*, as summarized in Table 3. Even the most informative descriptor, the 80th percentile of F_0 , attains an effect size of only $\eta^2 = 16.20\%$. The remaining features exhibit comparable behavior, thereby indicating that the classical acoustic parameters included in the *eGeMAPS* feature set provide only limited separability among these

highly activated emotional states.

These results provide motivation for the adoption of deep learning and representation learning approaches, such as Wav2Vec 2.0, which are capable of extracting higher-level features directly from raw audio. These models, explored in detail in the following chapters, have the potential to learn latent representations that encode subtler acoustic and temporal cues, thus improving discrimination among perceptually similar emotional states.

Speaker-dependent analysis of feature discriminativeness

The global analysis of acoustic features implicitly assumes that each characteristic is equally discriminative between speakers, an assumption that is rarely justified, because both speech production and emotional expression are influenced by physiological attributes and individual expressive strategies.

Consequently, features that appear highly discriminatory at the population level may provide limited discriminatory power for certain speakers, whereas features exhibiting only moderate separation in aggregate analysis may distinctly differentiate emotions for a specific speaker. This inter-speaker variability underscores the necessity of conducting speaker-specific analysis to capture individual expressive patterns that are obscured in population-level evaluations.

The following figures illustrate the speaker-dependent behavior of the mean of the fundamental frequency F_0 for the *Happy* and *Sad* speech. Although this feature shows a moderate separation at the population level, having Cohen's d equal to 1.816, its effectiveness is not always consistent when examined per-speaker.

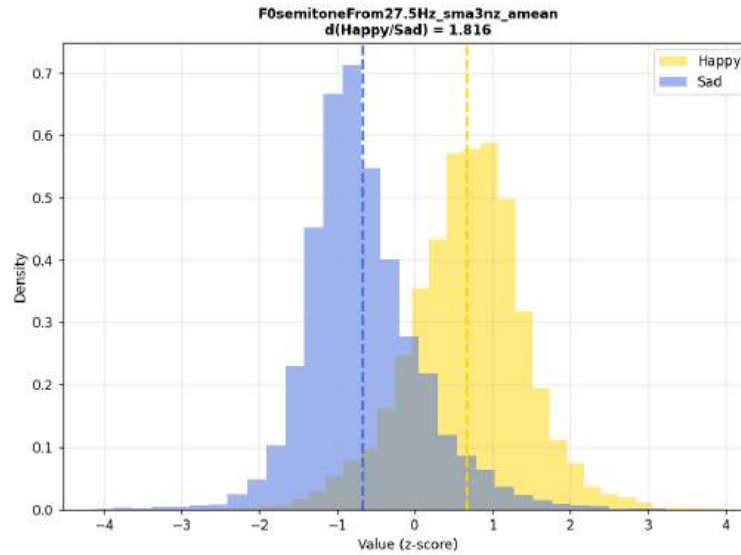


Figure 12: F_0 distribution of the *Happy* and *Sad* speech at the population level

For the speaker 0009, the distributions of happy and sad speech overlap considerably, indicating a limited discriminative utility for this feature. In contrast, speaker 0010 exhibits a markedly different pattern, having its distributions showing minimal overlap.

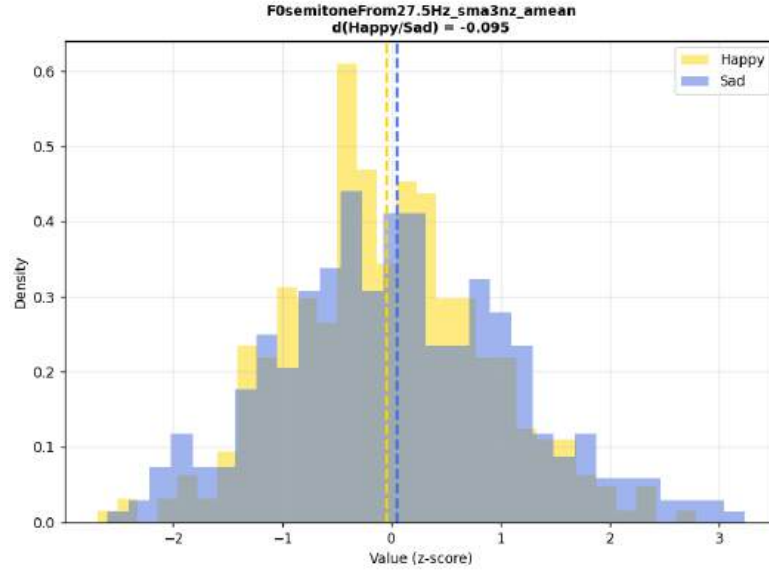


Figure 13: F_0 comparison for the *Happy* and *Sad* speech for speaker 0009

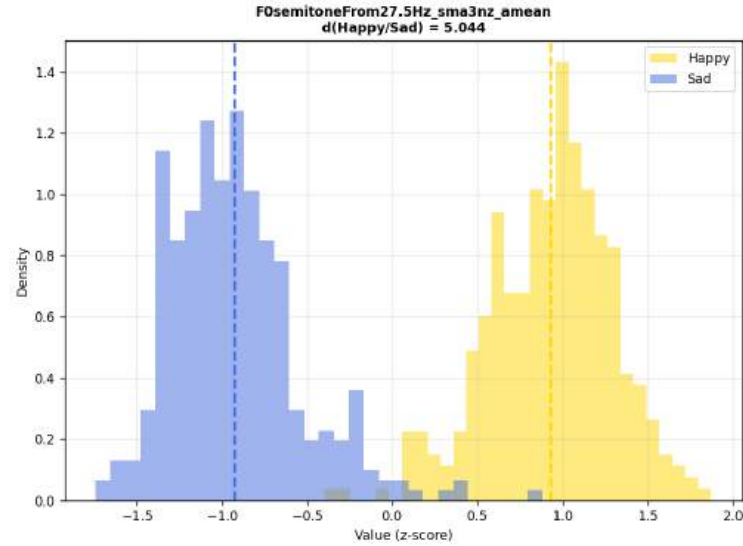


Figure 14: F_0 comparison for the *Happy* and *Sad* speech for speaker 0010

These observations show that the discriminative power of an acoustic feature does not generalize uniformly between speakers, even after normalization. Purely global feature-ranking methods may overestimate feature robustness by averaging across heterogeneous speakers, introducing the need for the use of speaker-aware emotion recognition systems.

The analysis of *equivalentSoundLevel_dBp* offers a complementary view of speaker dependent feature behavior, contrasting with the patterns observed for pitch-based features. At the population level, this energy-related feature shows only moderate discriminative capacity for the *Happy* and *Surprise* speech, having Cohen’s d equal to $|d| = 0.56$.

The measure *equivalentSoundLevel_dBp* indicates the average acoustic energy over time, expressed in decibels. The *Happy* and *Surprise* speech overlap largely along this dimension, suggesting overlapping local acoustic realizations of emotional categories for this speaker.

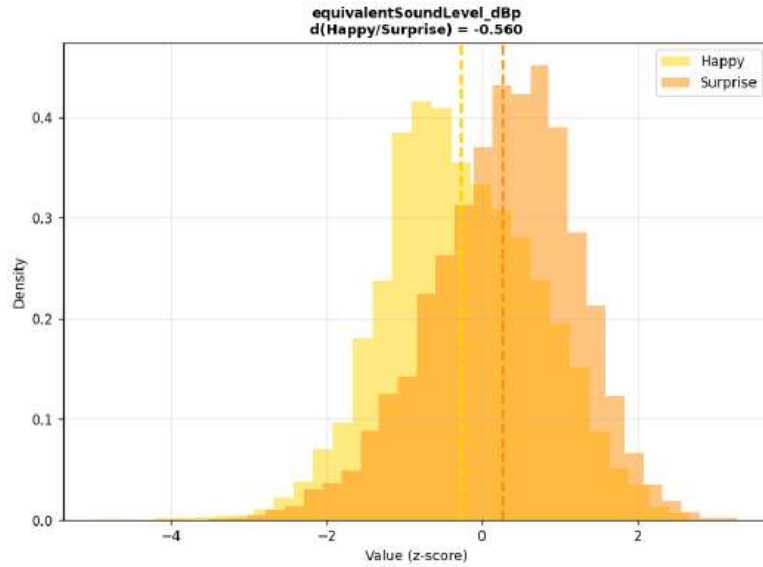


Figure 15: Loudness comparison between the *Happy* and *Surprise* speech

However, speaker-wise analysis might reveal substantial individual variation. For the speaker 0004, the *Happy* and *Surprise* speech are clearly separated by loudness, indicating that it is therefore an effective cue to differentiate between the two emotions ($|d| = 4.476$).

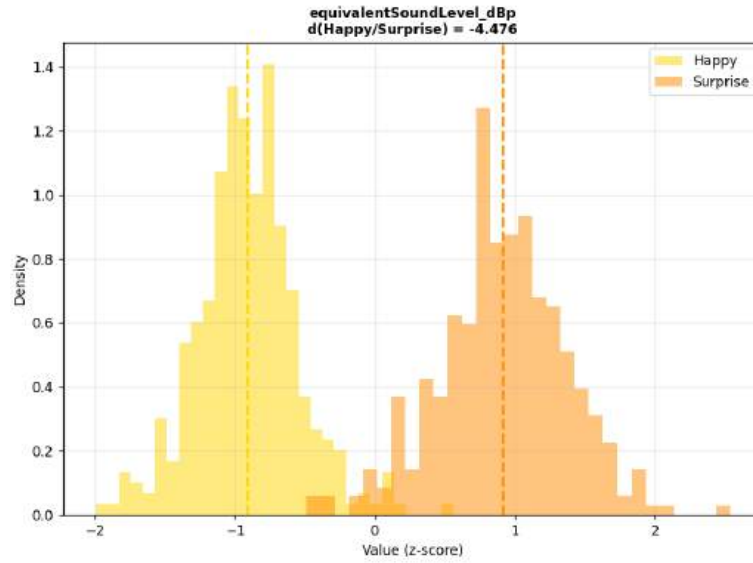


Figure 16: Loudness distribution for the *Happy* and *Surprise* speech for speaker 0004

These findings reinforce the importance of speaker dependent analysis in emotional speech research and show that discriminative strength varies between individuals. Previous research [28] has compared speaker dependent and speaker independent emotion recognition systems. Different models have been trained and tested on the same speaker versus models trained on multiple speakers and tested on unseen individuals. Across emotions, speaker-dependent systems achieved higher accuracy.

Chapter 3

Wav2Vec 2.0 and the structure of learned representations

This chapter introduces the architecture and underlying principles of the Wav2Vec 2.0 model, with an emphasis on its self-supervised learning framework [10, 29]. It describes how this approach represents a shift in learning speech representations and examines how the resulting embeddings are organized in the latent space, with the aim of associating physically meaningful acoustic properties to different emotional regions.

3.1 Self-supervised learning

Recent advances in artificial intelligence [30] have been largely driven by supervised learning approaches that rely on massive human-labeled datasets to detect patterns in data. Despite its success, this paradigm faces two major limitations.

Supervised models need large labeled datasets, and since human annotation is costly and slow, building high dimensional datasets is becoming increasingly impractical, resulting in limited scalability. Furthermore, supervised learning systems excel at narrowly defined tasks, but fail to transfer the acquired knowledge to related settings, resulting in poor generalization capabilities.

These issues drive a growing interest in self-supervised learning (SSL) as a complement to traditional supervision, because they leverage large unlabeled data for training and use labeled data in the final steps only for adaptation.

3.1.1 Contrastive learning

Contrastive learning [30, 31] is a self-supervised learning technique that aims to learn meaningful representations of unlabeled data by encouraging similarity between related samples and dissimilarity between unrelated ones.

Given an anchor input X , a corresponding positive example Y is expected to have a high similarity to X , while a set of negative examples $(Y'_1, \dots, Y'_K)$ should have a low similarity score. Since comparing an anchor with all possible negatives is computationally infeasible, a subset of negative samples is drawn for each anchor, exploration of the possibility Y is encouraged by a diversity loss function. The contrastive objective then forces the similarity between X and Y to be higher than the similarity between X and each sampled negative.

In natural language processing, BERT [30] provides a complementary example of self-supervised representation learning through its objective of masked language modeling, where parts of the input text are masked and the model is trained to predict missing tokens. For instance, inferring the masked word in a sentence such as “this is _ cat” requires exploiting the surrounding context, leading the model to implicitly learn syntactic and semantic regularities without relying on labeled data.

3.2 Discrete representations and quantization

Contrastive learning in natural language processing is based on the discrete and finite nature of the output space, where the models operate over a fixed vocabulary and produce probability distributions through a softmax layer. By contrast, to make contrastive learning applicable to audio, the signal must first be transformed into a discrete representation through discretization and quantization. In Wav2Vec 2.0, this process begins with a stack of feature encoders that generate multiple continuous latent representations that are then mapped to tokens by a quantization module.

Speech representations must encode both semantic and acoustic objectives [32]. In such settings, previous work on learning stable probability distributions during the contrastive sampling process has shown that optimization can admit degenerate solutions in which only a small subset of discrete units is effectively utilized, a phenomenon commonly referred to as index collapse [33].

The Wav2Vec 2.0 architecture explicitly addresses this issue through a quantization strategy based on the Gumbel-Softmax formulation, which is a differentiable approximation to discrete sampling (e.g., using the *argmax* function that produces a one-hot vector), enabling end-to-end training through backpropagation.

3.2.1 Gumbel-Softmax quantization

As stated above, the Gumbel–Softmax mechanism [34] provides a framework for differentiable sampling from categorical distributions. The procedure follows a four-stage pipeline.

Probability assignment The model first produces a categorical distribution over possible outcomes K , represented by probabilities $\pi_1, \pi_2, \dots, \pi_K$, where each π_i denotes the probability of selecting the category i .

Log-probability transformation The categorical probabilities are converted into logarithmic probabilities, yielding $\log(\pi_i)$ for each category.

Gumbel perturbation To enable exploration during training and to avoid deterministic selection of the maximum-probability category, each logarithmic probability is perturbed with a sample from the standard Gumbel distribution. For each category i , the noise G_i is drawn:

$$U_i \sim \text{Uniform}(0, 1), \quad G_i = -\log(-\log(U_i))$$

where $G_i \sim \text{Gumbel}(0, 1)$ having $\mu = 0$ and $\beta = 1$.

The perturbed logarithmic probabilities are then:

$$\log(\pi_i) + G_i$$

Temperature-scaled softmax Up to this stage, each category i is associated with a log-probability that has been perturbed by noise:

$$\log(\pi_i) + G_i.$$

If we were carrying out standard categorical sampling, we would simply pick the category with the largest value:

$$\arg \max_i (\log(\pi_i) + G_i).$$

However, the *argmax* operation is non-differentiable, which prevents its direct use in gradient-based optimization.

To overcome this, we substitute the *argmax* with a *softmax* function, which outputs a probability vector rather than a discrete one-hot vector:

$$y_i = \frac{e^{\frac{\log(\pi_i) + G_i}{\tau}}}{\sum_j e^{\frac{\log(\pi_j) + G_j}{\tau}}}$$

Temperature annealing is a training strategy in which a temperature parameter controls the sharpness of the output distribution. During training, the temperature follows an annealing schedule that starts from a relatively high value, producing smooth probability distributions and allowing exploration across classes. As training progresses, the temperature gradually reduces, leading to increasingly sharper distributions. In the final stages of training, the temperature becomes sufficiently low for the softmax output to closely approximate a one-hot vector, enabling the model to transition from exploratory behavior to nearly discrete decisions.

3.3 Wav2Vec 2.0 architecture

Wav2Vec 2.0 [1] is a self-supervised learning framework designed to learn informative representations of speech from raw audio signals. It relies on a contextual contrastive learning objective, where latent representations are learned by identifying the true quantized representation of masked time steps among a set of negatives, enabling the model to capture longer temporal dependencies.

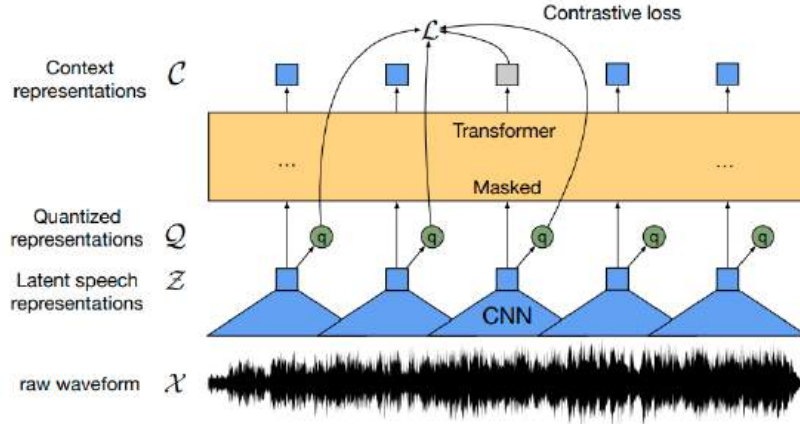


Figure 17: Wav2Vec 2.0 architecture. Figure source: A. Baevski, H. Zhou, A. Mohamed, and M. Auli, *wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations* [1].

3.3.1 The Wav2Vec 2.0 large XLSR variant

The reference model used in further analysis is *ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition* [35] obtained by fine-tuning a pretrained wav2vec 2.0 model on the RAVDESS dataset, to obtain an adaptation of general speech embeddings to the specific task of emotion recognition, resulting in an accuracy of 82.23%. In the next sections, the reference model is evaluated on the ESD dataset, which is composed of English and Chinese speech.

The RAVDESS dataset (*Audio-Visual Database of Emotional Speech and Song*) [36] comprises recordings of 24 professional actors (12 female and 12 male), all native English speakers with a neutral North American accent. The audio was recorded using a high-quality condenser microphone positioned, with a constant sampling rate of 48 kHz, and the audio levels were quantized to 16 bit.

The dataset includes a defined set of emotional categories for speech, consisting of $\{Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprise\}$ expressions. Except for neutral expressions, all emotions were produced at two explicitly controlled levels of emotional intensity, labeled normal and strong. The speech portion of the dataset contains a total of 1440 clips, balanced between actors, gender, emotional categories, and intensity levels.

This model is built on the multilingual wav2vec 2.0 XLSR-53 architecture [37], a large-scale speech representation model trained on $\approx 56,000$ hours of audio that span 53 languages, enabling the creation of cross-lingual speech representations.

This multilingual pretraining regime has an implication that is central to the subsequent chapters. Because XLSR-53 is trained contrastively on raw audio from 53 languages, with a quantization module that learns discrete units shared across them, it has a structural incentive to encode language as a distinct factor of variation in its representation [37]. Moreover, several of the languages used for evaluation in this thesis, most notably Mandarin, German, and French, are included in its pretraining corpus without any form of emotional annotation.

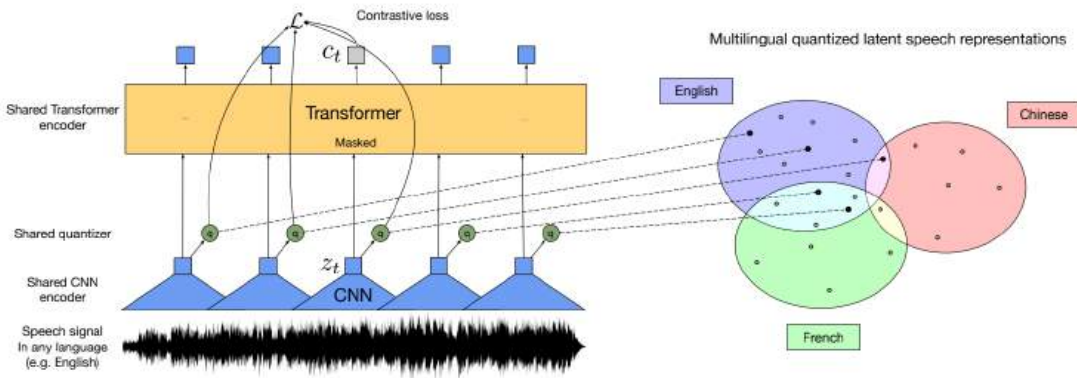


Figure 18: The XLSR-53 architecture. Figure source: A. Conneau, A. Baevski, R. Collobert, A. Mohamed, M. Auli, *Unsupervised Cross-lingual Representation Learning for Speech Recognition*.

Consequently, the cross-linguistic transfer effects reported in Chapter 5 should be interpreted as the transfer of emotional supervision across languages whose acoustic patterns the encoder has already encountered, rather than as generalization to languages that are entirely unseen by the encoder. This same characteristic also anticipates a key finding of the next chapter: an encoder that has learned to disentangle language as an independent factor is precisely one in which the emotion-related subspace can be approximately orthogonal to the language-related subspace.

The baseline model is composed of the following modules:

Feature encoder The first component of wav2vec 2.0 is the feature encoder, which is composed of several convolutional layers that transform the normalized raw waveform into a sequence of latent representations $\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{T'}\}$. Let

$$x \in \mathbb{R}^T$$

denote a raw audio signal sampled at frequency f_s , where T is the number of samples. The feature encoder implements a nonlinear mapping

$$f_{\text{enc}} : \mathbb{R}^T \rightarrow \mathbb{R}^{T' \times d},$$

where T' denotes the number of latent time steps after temporal downsampling and d denotes the dimensionality of the latent feature vectors. The output of the encoder is a sequence

$$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{T'}\}, \quad \mathbf{z}_t \in \mathbb{R}^d.$$

This mapping is realized through a stack of convolutional layers that progressively downsample the temporal dimension while increasing the representational capacity of the signal. Let s_i denote the stride of the i -th convolutional layer. The cumulative temporal downsampling factor is given by

$$D = \prod_i s_i,$$

which determines the length of the latent sequence as

$$T' = \left\lfloor \frac{T}{D} \right\rfloor.$$

In this model $d = 512$, the feature encoder thus produces a sequence of latent representations

$$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{T'}\}, \quad \mathbf{z}_t \in \mathbb{R}^{512}.$$

Each latent vector $\mathbf{z}_t$ summarizes the acoustic content of a local temporal segment of the original signal and serves as input to subsequent projection and contextual modeling stages.

```

Wav2Vec2ForSequenceClassification(
  (wav2vec2): Wav2Vec2Model(
    (feature_extractor): Wav2Vec2FeatureEncoder(
      (conv_layers): ModuleList(
        (0): Wav2Vec2LayerNormConvLayer(
          (conv): Conv1d(1, 512, kernel_size=(10,), stride=(5,))
          (layer_norm): LayerNorm((512,)), eps=1e-05, elementwise_affine=True)
          (activation): GELUActivation()
        )
        (1-4): 4 x Wav2Vec2LayerNormConvLayer(
          (conv): Conv1d(512, 512, kernel_size=(3,), stride=(2,))
          (layer_norm): LayerNorm((512,)), eps=1e-05, elementwise_affine=True)
          (activation): GELUActivation()
        )
        (5-6): 2 x Wav2Vec2LayerNormConvLayer(
          (conv): Conv1d(512, 512, kernel_size=(2,), stride=(2,))
          (layer_norm): LayerNorm((512,)), eps=1e-05, elementwise_affine=True)
          (activation): GELUActivation()
        )
      )
    )
    (feature_projection): Wav2Vec2FeatureProjection(
      (layer_norm): LayerNorm((512,)), eps=1e-05, elementwise_affine=True)
      (projection): Linear(in_features=512, out_features=1024, bias=True)
      (dropout): Dropout(p=0.05, inplace=False)
    )
    (encoder): Wav2Vec2EncoderStableLayerNorm(
      (pos_conv_embed): Wav2Vec2PositionalConvEmbedding(
        (conv): ParametrizedConv1d(
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          (parametrizations): ModuleDict(
            (weight): ParametrizationList(
              (0): _WeightNorm()
            )
          )
        )
      )
      (padding): Wav2Vec2SamePadLayer()
      (activation): GELUActivation()
    )
    (layer_norm): LayerNorm((1024,)), eps=1e-05, elementwise_affine=True)
    (dropout): Dropout(p=0.05, inplace=False)
    (layers): ModuleList(
      (0-23): 24 x Wav2Vec2EncoderLayerStableLayerNorm(
        (attention): Wav2Vec2Attention(
          (k_proj): Linear(in_features=1024, out_features=1024, bias=True)
          (v_proj): Linear(in_features=1024, out_features=1024, bias=True)
          (q_proj): Linear(in_features=1024, out_features=1024, bias=True)
          (out_proj): Linear(in_features=1024, out_features=1024, bias=True)
        )
        (dropout): Dropout(p=0.05, inplace=False)
        (layer_norm): LayerNorm((1024,)), eps=1e-05, elementwise_affine=True)
        (feed_forward): Wav2Vec2FeedForward(
          (intermediate_dropout): Dropout(p=0.05, inplace=False)
          (intermediate_dense): Linear(in_features=1024, out_features=4096, bias=True)
          (intermediate_act_fn): GELUActivation()
          (output_dense): Linear(in_features=4096, out_features=1024, bias=True)
          (output_dropout): Dropout(p=0.05, inplace=False)
        )
        (final_layer_norm): LayerNorm((1024,)), eps=1e-05, elementwise_affine=True)
      )
    )
  )
  (projector): Linear(in_features=1024, out_features=256, bias=True)
  (classifier): Linear(in_features=256, out_features=8, bias=True)
)

```

Listing 1: Architecture of the *ehcalabres/wav2vec2-lg-xlsr-en-speech-emotion-recognition* model

After the convolutional feature encoder produces latent speech representations, positional embeddings are added to encode temporal order, and the resulting representations are passed to the Transformer context network.

Feature projection The feature projection module maps convolutional features into the latent space expected by the Transformer encoder. Let

$$\mathbf{z}_t \in \mathbb{R}^{512}$$

denote the output of the convolutional feature encoder at the time step t . The projection is implemented as a linear transformation that yields latent representations

$$\tilde{\mathbf{z}}_t \in \mathbb{R}^{1024}.$$

Layer normalization is then applied to stabilize training by ensuring comparable activation scales across time steps, and dropout is applied to the normalized representations.

Quantization module The quantization module maps the latent representations to a discrete set of codes through the Gumbel-Softmax quantization mechanism explained in the previous section. These codes are then used during the contrastive phase. Let

$$\mathbf{z}_t \in \mathbb{R}^{512}$$

denote a latent representation produced by the feature encoder. The quantization module implements a mapping

$$Q : \mathbb{R}^{512} \rightarrow \mathcal{C},$$

where $\mathcal{C} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_K\}$ is a finite set of codebook vectors learned. For each time step t , the quantized representation is given by

$$\mathbf{q}_t = Q(\mathbf{z}_t),$$

which selects a discrete code from the codebook. These quantized vectors serve as target representations in the masked contextual contrastive learning objective.

To mitigate the index collapse problem and encourage codebook diversity, wav2vec 2.0 adopts a quantization scheme based on multiple smaller codebooks rather than a single large one [33]. Specifically, the model uses M smaller codebooks, each with N entries. During forward propagation, an entry is independently selected from each codebook and is then concatenated to form the final quantized representation.

Transformer encoder The core of the architecture is a Transformer encoder composed of 24 identical layers, which maps local acoustic features into globally contextualized speech representations. Each layer consists of a multi-head self-attention mechanism followed by a position-wise feed-forward network. Let

$$\mathbf{Z} = \{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{T'}\}, \quad \mathbf{z}_t \in \mathbb{R}^{1024},$$

denote the sequence of latent representations provided as input to the Transformer. During pre-training, a subset of these latent vectors is masked before being processed by the encoder. For each masked time step t , the Transformer produces a contextualized representation

$$\mathbf{c}_t = g(\mathbf{Z}_{\setminus t}),$$

where $g(\cdot)$ denotes the Transformer encoder and $\mathbf{Z}_{\setminus t}$ represents the input sequence with the masked t -th element. Due to the high correlation between consecutive frames of an audio signal, masking isolated positions would make the prediction trivial, as the model could simply copy information from nearby frames, thus a small proportion P of positions is selected as starting points, and from each of them, M consecutive time steps are masked.

The training objective encourages the contextual representation $\mathbf{c}_t$ to be close to the quantized target representation corresponding to $\mathbf{q}_t$, while being dissimilar from a set of k negative samples $\{\mathbf{q}'_k\}$, through a contrastive loss function.

Within each Transformer layer, the feed-forward network applies two linear transformations with a GELU nonlinearity, expanding the dimensionality from 1024 to 4096 and projecting it back to 1024. Layer normalization and dropout are applied throughout the encoder to ensure stable optimization and robust generalization.

Projection head The model applies a projection head to compress the Transformer outputs into a lower-dimensional latent space. Let

$$\mathbf{c}_t \in \mathbb{R}^{1024}$$

denote the contextualized representation produced by the Transformer encoder at the time step t . The projection head implements a linear mapping that yields a compact representation

$$\mathbf{h}_t \in \mathbb{R}^{256}$$

This reduction in dimensionality aims to concentrate information relevant to emotion in a compact latent space.

Classification layer For the specific task of SER, fine-tuning [38] can be performed by placing a classification layer on top of the pre-trained wav2vec 2.0 model and retraining it on

emotion-annotated datasets. This process refines the learned features to detect emotional patterns in spoken language.

The final component of the architecture is a linear classification layer that maps the projected representations to emotion scores. The classifier applies a linear mapping that produces a logit vector

$$\mathbf{o} \in \mathbb{R}^8$$

where each component corresponds to one of the eight emotion categories defined in the RAVDESS dataset. At inference time, a softmax function is applied to obtain a probability distribution over classes, and the predicted emotion is selected using the *argmax* function.

3.4 Geometry of the learned embeddings

Self-supervised speech representations define a high-dimensional space [39] in which distances reflect multiple latent factors of variation. The working hypothesis of this section is that, within a fixed speaker, local neighborhoods of this space may reflect acoustic and prosodic variations correlated with the emotional category of the utterance [40].

The analysis is organized in three steps: first, the extraction pipeline and the utterance-level pooling are described; then, a qualitative inspection based on t-SNE projections is conducted to motivate further analysis; finally, a k -nearest-neighbors (k -NN) probe is used to quantify the predictive value of the local structure.

3.4.1 Embedding extraction and utterance-level pooling

The embeddings analyzed in this chapter are extracted from the frozen backbone as follows. Each waveform is passed through the network, and only the *last* hidden state of the 24-layer Transformer encoder is retained: the convolutional feature encoder first segments the waveform into T' frames of dimension 512,

$$[T', 512],$$

which the feature projection maps to 1024 dimensions,

$$[T', 1024],$$

and the Transformer encoder then enriches each frame with contextual information from the surrounding time steps. The last-layer representation of an utterance is therefore a sequence of latent vectors

$$X = \{x_1, x_2, \dots, x_{T'}\}, \quad x_t \in \mathbb{R}^{1024},$$

or equivalently a matrix $X \in \mathbb{R}^{T' \times D}$, with $D = 1024$, where rows index temporal frames and columns index the latent dimensions of the last Transformer layer.

To obtain a fixed-length utterance-level representation, statistics pooling is applied along the temporal dimension. Mean pooling captures the central tendency of each latent dimension,

$$\mu_j = \frac{1}{T'} \sum_{t=1}^{T'} x_{t,j}, \quad j = 1, \dots, 1024,$$

while standard deviation pooling captures its temporal variability,

$$\sigma_j = \sqrt{\frac{1}{T'} \sum_{t=1}^{T'} (x_{t,j} - \mu_j)^2}, \quad j = 1, \dots, 1024.$$

The final utterance embedding is obtained by concatenating these first- and second-order statistics [41],

$$\mathbf{z} = [\mu \parallel \sigma] \in \mathbb{R}^{2048},$$

jointly encoding the average behaviour and the temporal variability of the latent features across the utterance. All subsequent analyses operate on the representation $\mathbf{z}$.

Statistics pooling retains complementary information by jointly aggregating the mean and standard deviation for each latent dimension. The mean, μ , reflects the central tendency of the latent representation across the utterance, while the standard deviation, σ , characterizes its temporal fluctuations. This second-order statistic thus yields an embedding-level encoding of the dynamic content. By comparison, relying solely on mean pooling collapses each latent trajectory to its centroid, thereby eliminating temporal variation.

3.4.2 Local structure via t-SNE projection

To inspect how utterance embeddings are locally organized, the high-dimensional vectors $\mathbf{z}$ are projected onto two dimensions via t-SNE [42]. It should be emphasized that this analysis is intentionally exploratory and visual in nature: t-SNE preserves local affinities at the expense of global geometry, and the observations reported below are not intended as claims about the geometry of the embedding space, but as visual prompts that motivate the quantitative analysis presented in the following subsection.

The 2D affinity structure varies substantially across speakers. For Speaker 0014, the projected embeddings appear largely intermingled across emotional labels, suggesting that, for this speaker, emotion is not the dominant factor governing local proximity in the learned representation.

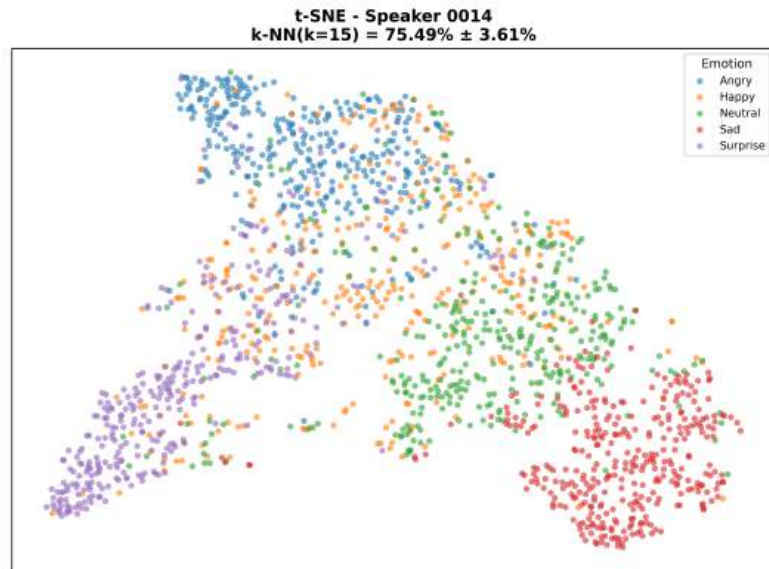


Figure 19: t-SNE projection of wav2vec 2.0 utterance-level embeddings for Speaker 0014. Local neighborhoods mix several emotional categories.

In contrast, Speaker 0008 exhibits a more structured organization, with local neighborhoods often dominated by a single emotional category, indicating stronger local separability of emotional labels for this speaker.

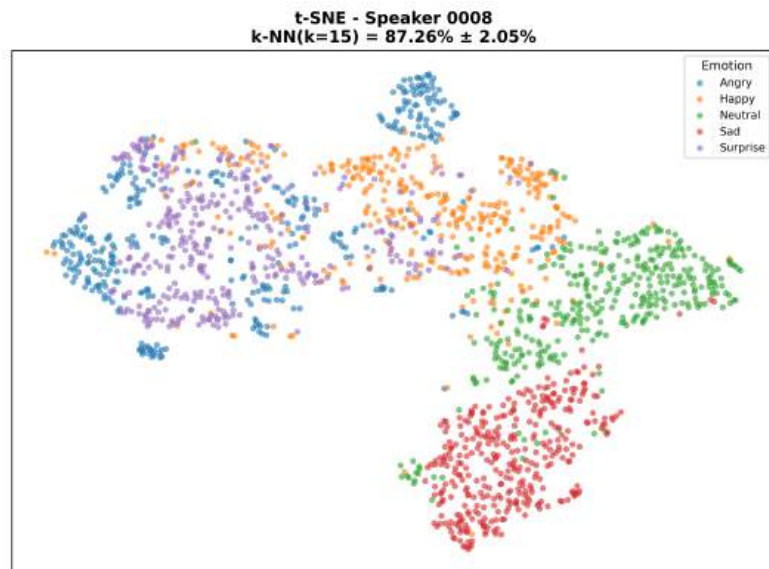


Figure 20: t-SNE projection of wav2vec 2.0 utterance-level embeddings for Speaker 0008. Local neighborhoods tend to be dominated by a single emotional category.

This contrast suggests that the legibility of emotion-related structure in the learned representation is itself speaker-dependent, and motivates a quantitative check of whether this local arrangement is in fact predictive of the emotional category.

3.4.3 Quantitative probing via k -nearest neighbors

To move from visual inspection to a measurable signal, a k -NN classifier is applied as a probe directly on the original 2048-dimensional utterance embeddings z . The rationale is that, if local neighborhoods in the embedding space carry information about the emotional category, a classifier that decides by analogy with nearby samples should perform appreciably above chance.

When training and testing are restricted to a single speaker, an average k -NN accuracy of $80.29\% \pm 7.21\%$ ($k = 15$) is obtained across speakers, indicating that emotion labels are largely predictable from local structure when speaker identity is held constant.

This per-speaker accuracy varies considerably, from 63.4% to 91.0%, mirroring quantitatively the speaker-dependent legibility already observed in the t-SNE projections of Speakers 0014 and 0008.

Table 4: Per-speaker k -NN accuracy on ESD embeddings ($k = 15$, speaker-internal cross-validation). Speakers 0001–0010 are Mandarin, 0011–0020 English.

Speaker	Accuracy (%)	Speaker	Accuracy (%)
0001	78.4 ± 3.3	0011	63.4 ± 2.7
0002	80.5 ± 2.4	0012	81.8 ± 3.2
0003	79.4 ± 2.5	0013	71.4 ± 4.1
0004	87.1 ± 4.3	0014	75.5 ± 3.6
0005	72.1 ± 2.2	0015	75.8 ± 3.8
0006	71.5 ± 3.9	0016	89.1 ± 2.1
0007	86.2 ± 3.4	0017	91.0 ± 2.2
0008	87.3 ± 2.0	0018	83.4 ± 1.4
0009	79.4 ± 1.8	0019	84.8 ± 2.1
0010	78.9 ± 3.2	0020	88.8 ± 2.6
Mean across speakers: 80.29 ± 7.21			

When the embeddings of all speakers are pooled and the probe is retrained on the unified set, the average accuracy drops to $70.04\% \pm 5.46\%$ ($k = 15$).

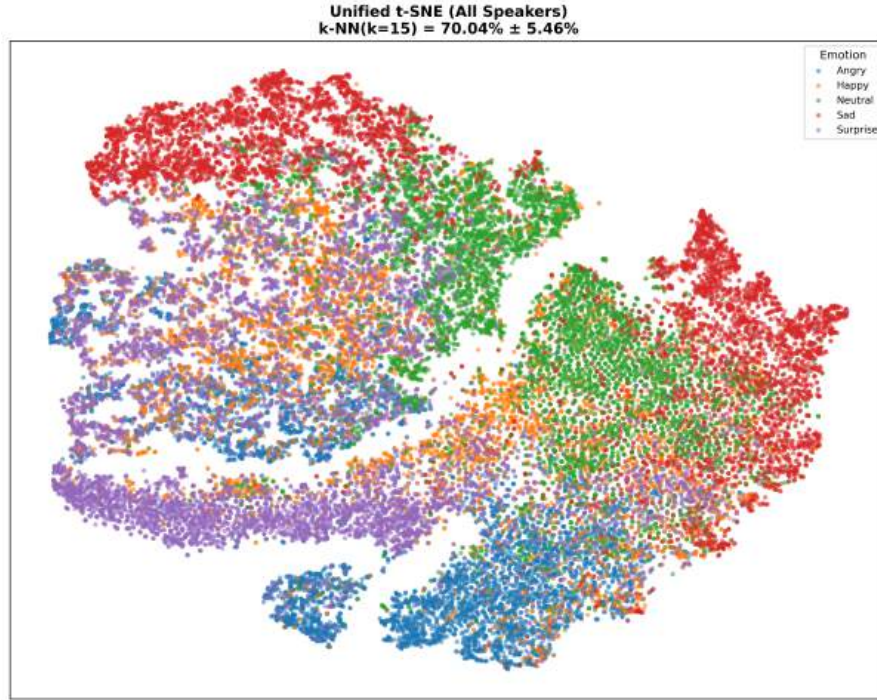


Figure 21: t-SNE projection of the pooled wav2vec 2.0 embeddings across all ESD speakers.

The gap between the within-speaker accuracy and the pooled-speaker accuracy, both measured at $k=15$, indicates that emotion-related structure coexists in the embedding space with other latent factors, which tend to dominate local proximity once multiple speakers are pooled.

The local probe of this section establishes that emotional information is present in the representation, but it does so only locally and from a within-speaker perspective: it neither isolates the factors that compete with emotion in governing the geometry, nor tells whether emotion remains separable once the representation is examined globally and across datasets.

These two limitations set the agenda for the next chapter, where the same frozen representation is studied from a global, dataset-level perspective. The analysis moves from local neighborhoods to the linear directions along which emotions are separated, asks how these relate to the directions encoding language, gender, and speaker identity, and compares them across datasets, in order to localize where a model trained on one corpus fails when transferred to another.

Chapter 4

A geometric diagnosis of cross-dataset transfer

The chapter is organized around a single guiding theme: localizing the origin of the cross-dataset failure within the model. Concretely, the speech emotion recognition model of [35], fine-tuned on RAVDESS, is evaluated on ESD without any further training or adaptation. Only the five emotion categories shared by the two corpora are considered: *Angry*, *Happy*, *Neutral*, *Sad*, and *Surprised*. The evaluation subset is class-balanced, with approximately 7,000 utterances per class (34,994 in total; six recordings were discarded as corrupted during extraction).

To localize the failure, the model is decomposed into its two main components: the encoder, which should provide a general acoustic representation of speech, and the classification head, whose decision boundaries are tied to the training-time label distribution. The goal is to determine which of the two is responsible for the loss of predictive performance on ESD. The remainder of the chapter formalizes this decomposition and uses linear probing to discriminate between the two possible sources of failure.

4.1 A cross-dataset transfer failure

The model fine-tuned by [35] on RAVDESS is evaluated on ESD without any further training or adaptation, restricted to the five emotion categories shared by the two corpora: *Angry*, *Happy*, *Neutral*, *Sad*, and *Surprised*. The ESD evaluation subset is class-balanced, with approximately 7,000 utterances per class. This section reports the model’s overall performance on the new dataset and inspects the structure of its errors, providing the empirical starting point for the rest of the chapter.

Table 5: Overall performance on ESD for the five shared emotion classes.

Metric	Value (%)
Accuracy	19.73
Macro Precision	21.28
Macro Recall	19.73
Macro F1-score	18.38

At first sight, the accuracy of 19.73% is indistinguishable from the 20% expected from uniform random guessing on a five-class problem. The per-class behaviour, however, is strongly non-uniform, as shown by the confusion matrix in Table 6.

Table 6: Confusion matrix of the baseline head on the ESD evaluation set. Rows are true labels, columns predicted labels.

	Angry	Happy	Neutral	Sad	Surprised
Angry	1667	4704	462	96	69
Happy	2780	2344	408	140	1327
Neutral	1592	2850	1510	13	1035
Sad	459	1388	896	74	4181
Surprised	2836	2334	28	492	1309

Three structural patterns emerge. First, the predictive mass concentrates on *Angry*, *Happy*, and *Surprised*: most utterances, regardless of true label, are assigned to one of these three classes, a collapse visible by column, with most rows feeding their mass into the *Angry* and *Happy* columns. Second, *Sad* is almost never predicted: only 74 of 7,000 truly sad utterances are correctly classified, the bulk being labelled *Surprised* instead. Third, the high-arousal classes *Angry*, *Happy*, and *Surprised* are systematically confused with one another.

4.2 Two competing hypotheses

To localize the source of this failure, the model can be decomposed into its two main components: the self-supervised speech encoder (wav2vec 2.0) and the supervised classification head fine-tuned on top of it. The encoder, pre-trained on a large amount of unlabelled speech, captures general acoustic properties of the signal, such as pitch, rhythm, energy, and spectral structure, that are tied to the physical mechanism of speech production rather than to any labelling scheme. The classification head, in contrast, was trained explicitly to associate regions of the encoder’s output space with the five RAVDESS emotion labels; the location and orientation of its decision boundaries are entirely determined by how these emotions are distributed in RAVDESS at training time.

This decomposition suggests two reference hypotheses for the observed failure:

- **H1 (representation hypothesis).** The encoder representation no longer organizes ESD utterances along emotion-relevant dimensions: emotional information is degraded or absent in the embedding space, so no classifier built on top of it could perform well.
- **H2 (alignment hypothesis).** The encoder representation still carries emotion-relevant information on ESD, but the decision boundaries learned on RAVDESS are misaligned with how emotions are distributed in the ESD embedding space, so a classifier that has not seen ESD cannot exploit it.

These describe two ends of a spectrum rather than a binary choice: the observed behaviour may combine them, and the aim of this chapter is not to certify one and rule out the other, but to locate where the failure predominantly originates and which component accounts for the larger share of the lost performance. Under a representation-dominated regime, any intervention would need to act on the encoder itself; under an alignment-dominated regime, the encoder can be left untouched and only the decision boundaries re-fitted.

The natural tool for this is linear probing: a linear classifier is trained directly on the ESD encoder embeddings and evaluated on a held-out portion of the same dataset. High probe accuracy would indicate that the embeddings linearly separate the emotional categories, ruling out the strong form of H1; accuracy close to chance would indicate that the embeddings themselves do not separate emotions, supporting H1. Intermediate accuracies quantify how much emotional structure survives in the target representation. The next section formalizes this setup and reports its results.

4.3 Linear probing of the embedding space

To discriminate between the two hypotheses, the quality of the embedding representation must be assessed independently of the classification head fine-tuned on RAVDESS.

Linear probing provides a principled way to do so [43, 44]: a new linear classifier is trained from scratch on the target dataset’s encoder embeddings, and its accuracy measures the linear separability of the classes in the representation. The probe assigns a label to each embedding $\mathbf{z} \in \mathbb{R}^d$ through

$$\hat{y} = \arg \max_c (\mathbf{w}_c^\top \mathbf{z} + b_c),$$

where $\mathbf{w}_c$, together with b_c , defines a class-specific decision hyperplane, and the inner product $\mathbf{w}_c^\top \mathbf{z}$ measures how strongly the embedding aligns with the direction associated with emotion c . High probe accuracy implies the classes occupy regions admitting a fixed set of separating hyperplanes [45, 46]; low accuracy implies no such global linear structure exists, and emotional information, if present, is encoded non-linearly.

A multinomial logistic regression classifier [47, 48] was therefore trained and independently evaluated on both datasets, using speaker-independent (GroupKFold) cross-validation on standardized 2048-dimensional embeddings. The RAVDESS task involved eight emotion classes (*Angry, Calm, Disgust, Fearful, Happy, Neutral, Sad, Surprised*); the ESD task involved the five shared classes from Section 4.1.

Table 7 reports the outcome of this probing protocol, trained and evaluated independently on each corpus.

Table 7: Logistic regression performance on speech emotion embeddings (%).

Metric	ESD	RAVDESS
Accuracy	77.64	92.15
Precision	77.50	91.90
Recall	77.64	91.67
F1-score	77.55	91.77

On RAVDESS, the linear probe reaches 92.15% accuracy, confirming that the encoder produces an internally consistent geometric organization of the emotions on its training corpus. On ESD, the same protocol reaches 77.64%, below RAVDESS but roughly 58 percentage points above the 19.73% obtained by the baseline head on the same data (Table 5). This aggregate figure, however, hides a non-trivial structure, reported per class in Table 8.

Table 8: Per-class classification report for logistic regression (speaker-independent GroupKFold cross-validation, %).

Class	Precision		Recall		F1-score	
	ESD	RAVDESS	ESD	RAVDESS	ESD	RAVDESS
Angry	81.29	95.34	84.57	95.83	82.90	95.58
Calm	–	89.34	–	91.67	–	90.49
Disgust	–	95.85	–	96.35	–	96.10
Fearful	–	95.70	–	92.71	–	94.18
Happy	63.25	89.12	61.65	89.58	62.44	89.35
Neutral	82.93	88.04	86.63	84.38	84.74	86.17
Sad	88.94	86.98	86.90	86.98	87.91	86.98
Surprised	71.08	94.85	68.48	95.83	69.76	95.34
Overall	77.50	91.90	77.64	91.67	77.55	91.77

Two contrasting regimes emerge. On RAVDESS, every class is recovered with F1 above 86%, confirming a uniformly structured emotional geometry. On ESD the picture is heterogeneous: *Angry, Neutral, and Sad* are separated well, with F1 in the 83%–88% range, whereas *Happy* and

Surprised fall to 62.44% and 69.76%. The confusion matrix in Table 9 localizes the difficulty: the two weak classes are confused primarily with *each other*, with 1,288 *Happy* utterances assigned to *Surprised* and 1,459 in the opposite direction, consistent with the high-arousal overlap already observed at the acoustic level in Chapter 2.

Table 9: Confusion matrix of the ESD linear probe (rows: true, columns: predicted). Correct counts on the diagonal in bold.

	Angry	Happy	Neutral	Sad	Surprised
Angry	5918	493	112	11	464
Happy	638	4315	465	293	1288
Neutral	128	339	6064	363	106
Sad	24	216	585	6081	92
Surprised	572	1459	86	89	4793

This heterogeneity already qualifies the diagnosis. The probe rules out the strong form of H1, since emotional information is not absent from the ESD representation: it is recovered at 77.64%, against the 19.73% of the transferred head. Yet the gap to the in-domain 92.15%, concentrated almost entirely on the two high-arousal classes, shows that the representation is not equally legible across the two corpora.

4.3.1 Linear encoding of non-emotional attributes

The probing experiment establishes that linear directions associated with emotion exist in the embedding space, but not that these directions are purely emotional. Self-supervised speech encoders are known to represent multiple latent factors of variation simultaneously, and emotion is only one of them. Before drawing further conclusions, it is therefore necessary to assess how strongly the same embedding space encodes other, non-emotional attributes of the signal. To this end, additional linear probes were trained under the identical speaker-independent (GroupKFold) protocol, targeting in turn the language, speaker identity, and gender of each utterance.

Table 10: Summary of linear probe results using a frozen wav2vec 2.0 backbone (%). All probes operate on standardized 2048-dimensional embeddings under speaker-independent (GroupK-Fold) cross-validation, except the speaker probe, which uses stratified cross-validation.

Attribute	Accuracy		Precision		Recall		F1	
	ESD	RAVDESS	ESD	RAVDESS	ESD	RAVDESS	ESD	RAVDESS
Language	99.84	–	99.84	–	99.84	–	99.84	–
Speaker	96.01	83.06	96.01	83.09	96.01	83.06	96.01	82.97
Gender	88.13	84.03	88.04	84.03	87.96	84.03	88.00	84.03
Emotion	77.64	92.15	77.50	91.90	77.64	91.67	77.55	91.77

The hierarchy is unambiguous: on ESD, language is almost perfectly recoverable (99.84%), followed by speaker identity (96.01%) and gender (88.13%), with emotion the least separable of the four attributes at 77.64%. On RAVDESS the ordering is inverted: emotion becomes the most separable attribute (92.15%), above gender (84.03%) and speaker (83.06%), because the encoder was explicitly fine-tuned on RAVDESS emotion labels, which structured the representation along emotion-aligned directions for that corpus.

The emotional information in the ESD embedding space is therefore real, but the same space encodes other attributes along more prominent linear directions. Probing accuracy alone does not reveal whether the linear separability of emotion reflects genuine affective structure or relies on its correlations with speaker, gender, and language. The following sections address this through explicit analyses of attribute entanglement and ablation.

4.4 Geometric alignment of latent factors

The previous section established that emotional information is linearly accessible in the ESD embedding space. This section extends the analysis along two complementary geometric directions. The first examines, *within* each dataset, whether the directions that separate emotions share geometric support with those associated with other prominent attributes. The second examines, across datasets, whether RAVDESS and ESD organize the same emotional categories along the same geometric axes. A single tool answers both, principal angles between subspaces [49, 50], and the two answers point in opposite directions, together revealing the geometric mechanism behind the transfer failure observed at the start of the chapter.

4.4.1 Geometric independence of emotion within each dataset

Each linear probe associates an attribute with a set of class-discriminative directions in the embedding space. Comparing the subspaces spanned by these directions across attributes determines whether they overlap or are approximately orthogonal [51].

Let $\mathbf{z} \in \mathbb{R}^d$ denote a latent embedding and $W \in \mathbb{R}^{C \times d}$ the weight matrix learned by a linear probe, where each row $\mathbf{w}_c \in \mathbb{R}^d$ corresponds to one class of the target attribute.

The emotion subspace is $\mathcal{E} = \text{span}(W_{\text{emo}})$, and the subspace of another attribute analogously $\mathcal{A} = \text{span}(W_{\text{att}})$. Multiclass subspaces are row-centered before the angle computation, removing the shared mean direction that does not discriminate between classes; a C -class attribute therefore spans a $(C-1)$ -dimensional subspace.

The geometric relationship between $\mathcal{E}$ and $\mathcal{A}$ is quantified through the principal angles $\{\theta_i\}$, defined by

$$\cos(\theta_i) = \sigma_i(U_{\mathcal{E}}^{\top} U_{\mathcal{A}}),$$

where $U_{\mathcal{E}}$ and $U_{\mathcal{A}}$ are orthonormal bases of the respective subspaces and $\sigma_i(\cdot)$ the singular values in decreasing order. The smallest principal angle θ_1 measures the closest possible alignment between any direction in the two subspaces: small values indicate shared geometric support, whereas values close to 90° indicate near-orthogonality and therefore geometric independence.

Table 11: Geometric alignment between the emotion subspace and the subspaces associated with non-emotional attributes, measured by the smallest principal angle between subspaces. The emotion subspace has dimension 4 on ESD and 7 on RAVDESS.

Comparison	Smallest principal angle		Subspace dim.	
	ESD	RAVDESS	ESD	RAVDESS
Emotion vs Language	85.03°	–	1	–
Emotion vs Gender	75.99°	79.82°	1	1
Emotion vs Speaker	75.10°	67.93°	19	23

In every comparison, the smallest principal angle exceeds 67° : the emotion subspace is therefore close to orthogonal to the subspaces of language, gender, and speaker identity in both datasets. The residual coupling is strongest with speaker identity, and slightly more pronounced on RAVDESS (67.93° , against actor identity) than on ESD (75.10°). This is the expected place for residual overlap: the speaker subspace is by far the highest-dimensional of the three, so it offers the most directions along which a small alignment with emotion can occur by chance.

4.5 Ablation of non-emotional subspaces

The previous section showed that the emotion subspace is approximately, but not exactly, orthogonal to the subspaces of language, gender, and speaker identity. The small residual overlap leaves open a quantitative question: whether the emotional separability measured by the linear probe relies, even partially, on the non-orthogonal component shared with these non-emotional attributes.

This section addresses it through a direct intervention. For each non-emotional attribute, the directions identified by the corresponding probe are assembled into a subspace, and the embeddings are mapped onto its orthogonal complement, removing every component along the probe’s directions while leaving intact those orthogonal to them. A new emotion probe is then trained from scratch on the projected embeddings and evaluated under the baseline protocol. If emotion accuracy drops after removal, part of the original separability was supported by the removed directions; if it remains unchanged, emotion is encoded along directions independent of those carrying the removed attribute. The procedure is a one-shot variant of Iterative Null-space Projection (INLP) [52].

4.5.1 Ablation results

The procedure is applied to the full ESD and RAVDESS embeddings under four conditions: removal of the language, gender, or speaker subspace, and of all three jointly. In each case, a new emotion probe is trained on the projected embeddings.

Table 12: Emotion probe performance before and after orthogonal removal of individual and combined non-emotional subspaces (%).

Condition	Accuracy		F1 (macro)	
	ESD	RAVDESS	ESD	RAVDESS
Remove Language	76.68	–	76.62	–
Remove Gender	76.60	91.81	76.54	91.41
Remove Speaker	77.25	92.01	77.17	91.63
Remove ALL nuisance	77.20	92.01	77.13	91.64

The pattern in Table 12 is consistent across both datasets. Removing any single non-emotional subspace changes emotion accuracy by about a percentage point in either direction, and removing all three jointly leaves accuracy essentially at baseline; the macro-averaged F1-score behaves identically.

Together with the principal-angle results, which established that the emotion subspace is approximately orthogonal to those of language, gender, and speaker identity, the ablation shows that the small residual overlap carries no measurable emotional signal: emotion classification on these embeddings does not exploit the directions associated with the non-emotional attributes.

4.6 Geometric alignment via Procrustes rotation

The analyses so far have established two facts. First, emotion is linearly separable in the embedding space of both RAVDESS and ESD, along directions essentially independent of those

carrying language, gender, and speaker identity. Second, this within-dataset decodability does not imply that an identical linear mapping generalizes across datasets.

A logistic regression probe learns, for each emotion class c , a weight vector $\mathbf{w}_c \in \mathbb{R}^d$ defining the direction that best separates class c from the others; classification reduces to a projection, since the larger $\mathbf{w}_c^\top \mathbf{z}$, the more strongly embedding $\mathbf{z}$ is associated with emotion c . The RAVDESS and ESD probes can each achieve high in-domain accuracy while relying on different directions. When these directions align across datasets, a classifier trained on one domain generalizes to the other, since the underlying emotion geometry is consistent; when they diverge, the same classifier projects embeddings from the unseen domain onto directions that are uninformative or misleading in the target space.

This section tests that interpretation directly. If the discrepancy between the two sets of directions is, to a good approximation, a difference of orientation, then a single rigid transformation should both align them and recover the accuracy lost when a baseline probe is applied to ESD without adaptation.

4.6.1 The Procrustes alignment problem

Procrustes analysis [53] finds the orthogonal transformation that best aligns one configuration of vectors with another. Here, the two configurations are the matrices of emotion-separating directions learned independently by the RAVDESS and ESD probes.

Let $\hat{W}^{rav}, \hat{W}^{esd} \in \mathbb{R}^{K \times d}$ denote the row-normalized weight matrices of the two probes, where K is the number of shared emotion classes and d the embedding dimensionality. Each row is the direction along which one emotion is optimally separated from the others in the respective dataset, and the cosine similarities in Table 13 confirm that corresponding rows of $\hat{W}^{rav}$ and $\hat{W}^{esd}$ are nearly orthogonal.

The goal is a rigid transformation of the embedding space that, applied to $\hat{W}^{rav}$, brings each row as close as possible to the corresponding row of $\hat{W}^{esd}$. Such a transformation is an orthogonal matrix $R \in \mathbb{R}^{d \times d}$ satisfying $R^\top R = I_d$, a constraint that restricts the search to maps preserving the internal geometry of the configuration, i.e. rotations and reflections. The Procrustes problem is then

$$R^* = \arg \min_{R^\top R = I_d} \left\| \hat{W}^{rav} R - \hat{W}^{esd} \right\|_F$$

where the residual matrix $\hat{W}^{rav} R - \hat{W}^{esd}$ collects, row by row, the discrepancies between each rotated RAVDESS direction and the corresponding ESD direction, and $\| \cdot \|_F$ is the Frobenius norm.

The minimization admits a closed-form solution through the singular value decomposition of the cross-covariance between the two configurations:

$$U \Sigma V^\top = \text{SVD}\left(\left(\hat{W}^{rav}\right)^\top \hat{W}^{esd}\right),$$

where $U, V \in \mathbb{R}^{d \times d}$ are orthonormal and Σ diagonal with non-negative entries. The matrix $(\hat{W}^{rav})^\top \hat{W}^{esd}$ encodes the alignment between each RAVDESS and each ESD direction, and the SVD identifies the pairs along which the two configurations are most strongly correlated. The optimal rotation is

$$R^* = U V^\top.$$

4.6.2 Alignment of emotion directions

To measure the alignment between the emotion directions of RAVDESS and ESD, the cosine similarity [50] between the corresponding weight vectors is computed for each of the five shared emotion classes.

Let $\mathbf{w}_c^{\text{RAV}}$ and $\mathbf{w}_c^{\text{ESD}}$ denote the weight vectors learned for class c by the RAVDESS and ESD probes, respectively. The cosine similarity is defined as follows:

$$\cos(\mathbf{w}_c^{\text{RAV}}, \mathbf{w}_c^{\text{ESD}}) = \frac{(\mathbf{w}_c^{\text{RAV}})^\top \mathbf{w}_c^{\text{ESD}}}{\|\mathbf{w}_c^{\text{RAV}}\| \|\mathbf{w}_c^{\text{ESD}}\|} \quad (3)$$

Table 13: Per-class cosine similarity between the RAVDESS and ESD emotion-separating directions, before and after the Procrustes rotation.

Emotion	Before	After
Angry	0.018	0.991
Happy	0.029	0.995
Neutral	0.042	0.985
Sad	0.045	0.990
Surprised	0.012	0.980

Table 13 reports the per-class cosine similarity before and after the rotation. Every emotion moves from near-orthogonal disagreement (cosine ≈ 0) to near-perfect agreement (cosine ≥ 0.98). These facts suggest that the encoder represents emotional information consistently across datasets, but expresses it along different geometric axes in each one.

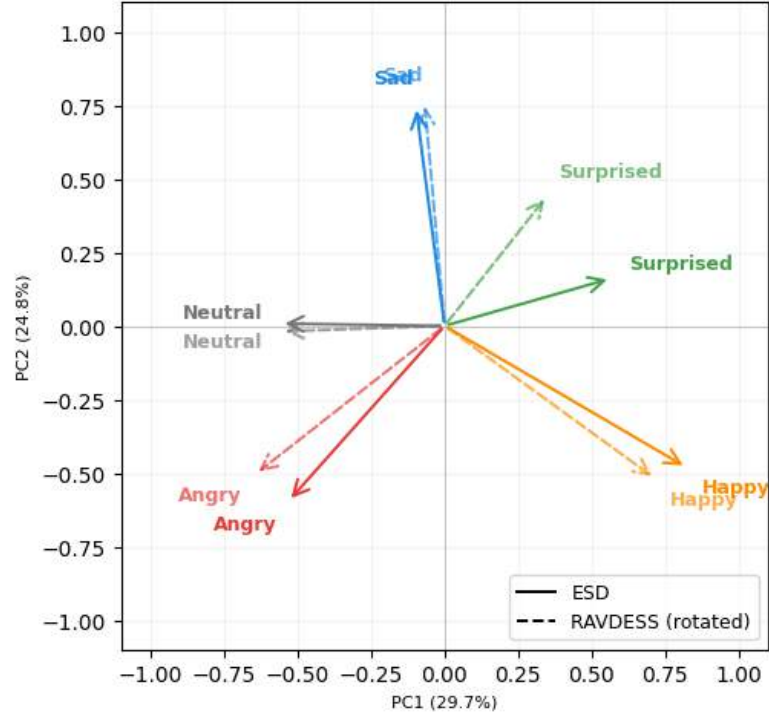


Figure 22: Emotion-separating directions of RAVDESS after Procrustes rotation, overlaid on the ESD directions, projected onto the first two principal components of the combined weight matrix.

4.6.3 Performance recovered by the rotation

A high post-rotation cosine establishes only that an aligning rotation *exists*; it does not establish that the rotation recovers emotional information. The decisive test is whether the rotation, estimated and evaluated on *disjoint* speakers, restores the classification accuracy the RAVDESS probe loses on ESD: a rotation that helped only on the data used to estimate it would be uninformative.

The validation is therefore speaker-independent. The RAVDESS probe is trained once and frozen. The ESD speakers are split into a fit set of 16 and a held-out set of 4, and three quantities are compared on the held-out speakers, who take no part in estimating anything:

- **Baseline.** The frozen RAVDESS probe classifies the held-out embeddings directly, with no rotation, measuring the unaided cross-dataset transfer.
- **Procrustes (rotated).** The rotation R^* is estimated from the fit speakers only; the held-out embeddings are mapped into the RAVDESS frame through R^* and classified by the

same frozen probe, measuring the transfer recovered by a rotation learned on *other* speakers.

- **Oracle.** An ESD-native probe, trained on the same fit speakers, classifies the held-out speakers directly: the in-domain reference ceiling.

The procedure is repeated over five random speaker splits; all quantities are reported as mean $\pm$ standard deviation.

Table 14: Speaker-held-out validation on ESD (4 held-out speakers, 5 random splits, mean $\pm$ std). The Procrustes rotation is estimated from the fit speakers only and applied to the held-out speakers, who are then classified by the frozen RAVDESS probe.

Condition	Accuracy (%)	Macro F1 (%)
Baseline (no rotation)	41.9 ± 2.1	37.8 ± 1.5
Procrustes (rotated)	71.0 ± 2.1	70.5 ± 2.8
Oracle (ESD-native probe)	75.2 ± 3.5	75.2 ± 3.6

The baseline reaches 41.9%: above the 20% chance level, but well below a usable accuracy. The rotation raises this to 71.0%, against an oracle of 75.2%. Since the held-out speakers do not contribute to estimating R^* , the recovery cannot be attributed to overfitting: the rotation learned on one group of speakers transfers the emotional geometry to a disjoint group.

Figure 23 shows the effect qualitatively: without rotation the predictions collapse onto a few columns, notably *Surprised*, and *Neutral* is never predicted; after rotation the matrix is diagonally dominant and every class is recovered.

(a) Baseline (no rotation)						(b) Procrustes (rotated)					
	A	H	N	S	Su		A	H	N	S	Su
A	618	288	0	8	486	A	1118	128	99	8	47
H	117	486	0	41	756	H	225	1035	49	46	45
N	90	633	2	191	484	N	31	74	1156	137	2
S	20	117	2	923	334	S	21	43	159	1173	4
Su	13	134	0	150	1103	Su	138	757	9	12	484

Figure 23: Confusion matrices for one held-out split (ESD speakers 0002, 0009, 0013, 0014), before (left) and after (right) the Procrustes rotation, using the frozen RAVDESS probe on unseen ESD speakers. Rows are true classes, columns predictions; correct counts in bold.

z

The Procrustes rotation serves here as a diagnostic tool, not an adaptation method: it characterizes the cross-dataset failure rather than correcting it. Emotional information is not lost on

ESD; it remains linearly accessible, and the directions that separate emotions differ from the RAVDESS ones mainly by an orientation that a transformation can correct.

This has a direct implication. If the obstacle is largely a rotational misalignment, a new classification head trained on the frozen encoder with ESD labels should absorb that rotation implicitly during gradient descent, without ever computing R^* . The next section pursues this strategy and measures how closely a small trained head approaches the performance recovered here by the rotation.

Chapter 5

From supervised recovery to zero-shot cross-corpus transfer

The diagnostic chapter showed that the baseline’s cross-dataset failure is a rotational misalignment, not a loss of emotional information in the encoder, and predicted that a head retrained on the frozen backbone should recover the lost performance. This chapter tests that prediction in two stages.

The first retrains a compact head on the frozen wav2vec 2.0 backbone with ESD labels, and shows it recovers near-probe accuracy. The second extends the test to three corpora absent from both supervised stages, in German, French, and Greek, languages that received no emotional supervision at any point. If the head captured emotion in a corpus-independent way, it should label these utterances correctly in pure zero-shot inference.

5.1 Head retraining on the frozen encoder

The diagnostic analysis showed that emotion remains linearly accessible in the ESD representation and that the baseline’s failure stems from a rotational misalignment a single transformation can largely correct. This motivates a direct strategy: rather than computing the rotation explicitly, a new classification head is trained on the frozen wav2vec 2.0 backbone with ESD labels, letting gradient descent absorb the misalignment implicitly.

5.1.1 Model architecture

The backbone, wav2vec 2.0 fine-tuned on RAVDESS, is kept frozen throughout: its parameters are never updated, so every claim here concerns only what a classifier can recover from an unchanged representation [54]. Each utterance is mean-pooled over time into a single embedding

$\mathbf{z} \in \mathbb{R}^{1024}$.

The head is deliberately compact. A learnable Layer Normalization approximates the global standardization under which the linear probes reached 77.64% on ESD; the result is projected to 256 dimensions, normalized again, and passed through a two-layer GELU MLP with dropout ($p = 0.1$), producing logits over the five shared emotions:

$$\mathbf{z}_{\text{norm}} = \text{LayerNorm}(\mathbf{z}), \quad (4)$$

$$\mathbf{h}_1 = \text{LayerNorm}(W_{\text{proj}} \mathbf{z}_{\text{norm}} + b_{\text{proj}}), \quad (5)$$

$$\mathbf{h}_2 = \text{Dropout}(\text{GELU}(\mathbf{h}_1)), \quad (6)$$

$$\mathbf{h}_3 = \text{Dropout}(\text{GELU}(W_2 \mathbf{h}_2 + b_2)), \quad (7)$$

$$\hat{\mathbf{y}} = W_3 \mathbf{h}_3 + b_3, \quad (8)$$

with $W_{\text{proj}} \in \mathbb{R}^{256 \times 1024}$, $W_2 \in \mathbb{R}^{128 \times 256}$, and $W_3 \in \mathbb{R}^{5 \times 128}$.

5.1.2 Training protocol

The head is trained on the full ESD set (all twenty speakers) under five-fold speaker-independent (GroupKFold) cross-validation, with an 80/20 train/validation split at the speaker level: sixteen training and four validation speakers per fold. The folds are stratified by language, so each validation fold holds exactly two English and two Chinese speakers. Batches are class-balanced so that each emotion is represented roughly equally. Training runs for up to 15 epochs with early stopping (patience 4), using cross-entropy with label smoothing ($\epsilon = 0.05$).

5.1.3 Recovering emotion classification on ESD

The first question is whether a head trained on the frozen encoder recovers the performance lost in the cross-dataset transfer. Table 15 reports the re-trained head against the two reference values established earlier.

Table 15: Overall accuracy and F1 on ESD of the re-trained head, compared against the two reference values established earlier. Mean and standard deviation across five speaker-independent (GroupKFold) folds.

Condition	Accuracy (%)	F1 (%)
Baseline model	19.73	18.38
Linear probe on ESD embeddings	77.64	77.55
Retrained head	75.12 ± 3.40	75.00 ± 3.70

The head recovers almost all of the lost performance: from the 19.73% of the original RAVDESS-trained head on the same evaluation set up to 75.12%, within roughly three points of the 77.64% linear-probe bound. Trained only on ESD labels, gradient descent therefore reaches an effect close to the Procrustes rotation characterized earlier in closed form. The recovery is stable across folds, as Table 16 shows.

Table 16: Retrained head: per-fold results under language-stratified cross-validation.

Fold	Val speakers (EN / ZH)	Best epoch	Accuracy (%)	F1 (%)
1	18, 20 / 7, 9	13	79.11	79.49
2	12, 16 / 8, 10	5	78.21	78.31
3	11, 17 / 3, 4	7	72.81	72.74
4	13, 14 / 1, 5	9	69.93	69.34
5	15, 19 / 2, 6	15	75.51	75.11
Mean $\pm$ std			75.12 $\pm$ 3.40	75.00 $\pm$ 3.70

Because each fold mixes English and Chinese validation speakers, this single head already operates across both languages; its overall per-class recall is reported in Table 17.

Table 17: Mean per-class recall (%) of the re-trained head on ESD, across the five speaker-independent folds.

Emotion	Mean recall (%)	Std
Angry	78.2	7.8
Happy	60.3	14.9
Neutral	90.5	6.4
Sad	80.1	5.5
Surprised	66.5	13.2

The head reads emotion across both languages, but not uniformly across classes. *Neutral* is recovered almost perfectly, whereas high-arousal states such as *Happy* and *Surprised* show lower recall and higher variance.

5.2 Evaluation corpora

Three acted emotional speech corpora are used for the zero-shot evaluation, selected to span three different languages: the Berlin Database of Emotional Speech (EmoDB) [55], the Canadian French Emotional Speech corpus (CaFE) [56], and the Acted Emotional Speech Dynamic Database (AESDD) [57].

EmoDB. EmoDB [55] is a German corpus recorded in the anechoic chamber of the Technical University of Berlin: 535 utterances from ten actors (five female, five male) over seven emotions (anger, boredom, disgust, fear, happiness, sadness, neutral), sampled at 16 kHz mono. Four overlap with the thesis label set, *Angry*, *Happy*, *Neutral*, and *Sad*, yielding 339 utterances.

CaFE. CaFE [56] is a Canadian French corpus recorded by twelve actors (six female, six male) over seven emotions (anger, disgust, happiness, fear, surprise, sadness, neutral), each uttering six semantically neutral sentences. All emotions except neutral are acted at two intensity levels, *faible* (weak) and *fort* (strong); neutral is recorded once. Five overlap with the thesis label set, *Angry*, *Happy*, *Neutral*, *Sad*, and *Surprised*, yielding 648 utterances.

AESDD. AESDD [57] is a Greek corpus recorded by professional actors over five emotions (anger, disgust, fear, happiness, sadness). With no neutral or surprised category, only three overlap with the thesis label set, *Angry*, *Happy*, and *Sad*, yielding 362 utterances.

5.3 Experimental setup

All evaluations are carried out in pure zero-shot inference: neither model is trained, fine-tuned, or adapted in any way on the three corpora.

The first model is the original baseline wav2vec 2.0 fine-tuned on RAVDESS, loaded with its eight-class head. Since this head produces logits over emotions outside the shared label set, the non-shared indices are masked before arg max, constraining predictions to the emotions present in each target corpus for a fair comparison.

The second is the frozen wav2vec 2.0 backbone paired with the head retrained on all twenty ESD speakers, with weights taken from the best-performing fold of that experiment. Since this head produces logits over five classes, the indices for emotions absent from a given corpus are likewise filtered out before evaluation.

5.4 Results

Table 18 reports the overall zero-shot performance of the two models on the three corpora. The retrained head improves over the RAVDESS baseline on every corpus and metric, though both the size of the improvement and the absolute level reached vary substantially.

Table 18: Zero-shot evaluation on the three unseen corpora. All values are percentages; precision, recall and F1 are macro-averaged over the shared classes of each corpus. The *Improvement* row reports the absolute gain of the retrained head over the base model.

Corpus	Model	Accuracy	Precision	Recall	F1
EmoDB	Base model	21.24	12.80	24.57	16.48
	ESD-retrained head	94.69	94.55	95.00	94.63
	Improvement	+73.45	+81.75	+70.43	+78.15
AESDD	Base model	61.16	52.50	60.64	49.49
	ESD-retrained head	74.93	83.21	75.02	75.00
	Improvement	+13.77	+30.71	+14.38	+25.51
CaFE	Base model	14.04	10.10	19.17	13.23
	ESD-retrained head	63.43	62.50	61.53	60.84
	Improvement	+49.39	+52.40	+42.36	+47.61

The ESD-retrained head reaches 94.69% on EmoDB, 74.93% on AESDD, and 63.43% on CaFE, all on languages absent from both the backbone fine-tuning and the head training.

The RAVDESS-trained head does poorly. On EmoDB and CaFE its accuracy (21.24% and 14.04%) falls below chance level, showing that its decision boundaries, fitted to the RAVDESS geometry, are misaligned with the unseen corpora. On AESDD it reaches 61.16%, above the three-class chance level, but only because a strong bias towards *Angry* and *Sad* absorbs nearly all predictions while *Happy* collapses entirely, as the per-class analysis below shows.

5.4.1 Per-class analysis

Table 19: Per-class recall (%) of the two models on the three corpora.

Corpus	Emotion	Base model	ESD head	Delta
EmoDB (339)	Angry	0.00	94.49	+94.49
	Happy	70.42	94.37	+23.95
	Neutral	27.85	91.14	+63.29
	Sad	0.00	100.00	+100.00
AESDD (362)	Angry	85.95	51.24	-34.71
	Happy	0.84	94.96	+94.12
	Sad	95.12	78.86	-16.26
CaFE (648)	Angry	0.00	84.72	+84.72
	Happy	25.00	36.81	+11.81
	Neutral	65.28	44.44	-20.84
	Sad	0.00	89.58	+89.58
	Surprised	5.56	52.08	+46.52

The baseline fails differently on each corpus. On EmoDB and CaFE it never predicts *Angry* or *Sad*, putting most mass on *Happy* and *Neutral*; on AESDD it does the opposite, predicting mostly *Angry* and *Sad* while *Happy* falls below 1%.

The retrained head recovers every class everywhere, though unevenly. *Sad* stays above 78% on all three corpora and *Happy* is strong on EmoDB and AESDD; CaFE is the weakest, with *Happy* and *Neutral* still hard to separate. The drops in *Angry* and *Sad* recall on AESDD are not real losses: the head stops over-predicting those classes, raising their precision and recovering *Happy* from near-zero to 94.96%.

5.4.2 Confusion structure

Table 20: Confusion matrices on the three unseen corpora. Top row: base RAVDESS model; bottom row: ESD-retrained head. Rows are ground-truth classes, columns predictions; correct counts (the diagonal) in bold. One AESDD utterance failed to load and is excluded.

(a) EmoDB - Base model					(b) AESDD - Base model				(c) CaFE - Base model					
A	H	N	S		A	H	S		A	H	N	S	Su	
A	0	63	5	59	A	104	2	15	A	0	113	13	0	18
H	3	50	17	1	H	48	1	70	H	0	36	31	1	76
N	0	57	22	0	S	5	0	117	N	0	9	47	7	9
S	5	0	57	0					S	0	4	17	0	123
									Su	0	104	32	0	8

(d) EmoDB - ESD-retrained					(e) AESDD - ESD-retrained				(f) CaFE - ESD-retrained					
A	H	N	S		A	H	S		A	H	N	S	Su	
A	120	7	0	0	A	62	58	1	A	122	10	0	4	8
H	3	67	0	1	H	2	113	4	H	16	53	12	35	28
N	0	6	72	1	S	0	25	97	N	6	2	32	31	1
S	0	0	0	62					S	0	8	4	129	3
									Su	27	38	2	2	75

The confusion matrices show that the baseline does not collapse onto a single class but onto a shifting subset, always sparing the high-arousal confusions: it spreads between *Happy* and *Neutral* on EmoDB, *Happy* and *Surprised* on CaFE, and *Angry* and *Sad* on AESDD, while the classes outside each bias drop to zero. The retrained matrices, by contrast, are diagonally dominant on all three corpora, the only substantial off-diagonal mass being the *Happy–Surprised* and *Happy–Sad* overlaps in CaFE, consistent with the high-arousal confusions seen earlier.

Chapter 6

Conclusions and future work

This thesis addressed the cross-corpus failure of a fine-tuned wav2vec 2.0 model for speech emotion recognition, following a single line of reasoning from diagnosis to intervention to evaluation.

The diagnostic chapters localized the failure. When the baseline model, fine-tuned for SER on RAVDESS, is applied to another corpus, emotional information remains linearly accessible in the encoder representation. The failure lies instead in the classification head, whose decision boundaries are tied to the geometry of the training corpus and become misaligned with that of any other.

The constructive part turned this diagnosis into a minimal intervention: if only the head is misaligned, retraining a small head on target-domain labels, with the encoder frozen, should suffice to recover performance. This was confirmed: a compact head trained on ESD over the frozen backbone restored emotional discrimination without modifying the encoder.

The previous chapter completed the arc with a zero-shot evaluation. The head trained only on RAVDESS and ESD was applied, without further adaptation, to three emotional speech corpora in languages absent from both supervised stages, improving over the baseline on every one.

6.1 From discrete labels to acoustic descriptors

The analyses supporting these conclusions describe the encoder representation almost entirely through *discrete labels*. Separability was characterized at a *macro* level, through the geometry of the emotion, language, gender, and speaker subspaces, each built by assigning a categorical label to every embedding. This view establishes *that* emotions are separable, and largely independently of non-emotional factors, but says nothing about *what* in the signal the encoder relies on.

A complementary perspective inspects the latent space through continuous acoustic descriptors. Rather than asking which categorical region an embedding belongs to, one can ask how interpretable acoustic and prosodic quantities, captured here by the *eGeMAPS* set, activate across the representation. Overlaying these descriptors on the embedding space replaces the categorical question (*which emotion?*) with an acoustic one (*which prosodic and spectral properties organize this region?*), linking the encoder geometry to physically interpretable dimensions of the signal.

The exploratory *eGeMAPS* analysis of Chapter 2 examined such descriptors in isolation and found high within-emotion variance and limited, often speaker-dependent, discriminative power: studied one feature at a time and detached from the representation, they are weak predictors. The question here is whether the same descriptors become meaningful once read *on* the encoder geometry rather than on the raw signal. The single-speaker case study below takes a first step in this direction.

6.1.1 A single-speaker case study

To gain intuition about what local proximity in the embedding space reflects, a feature-driven inspection overlays low-level *eGeMAPS* descriptors on the t-SNE projection of a single speaker. Speaker 0008 is chosen because its projection shows the clearest local organization (Figure 20); the analysis is therefore illustrative rather than representative of the dataset. For each descriptor, a heatmap is obtained by computing, at every point of the 2D grid, the average feature value of the $k = 5$ nearest projected utterances.

The three descriptors below fall into two families: *spectral flux* and *loudness falling slope* characterize the **temporal dynamics of energy**, while *alpha ratio* characterizes the **shape of the spectral envelope**.

Spectral flux measures how rapidly the spectral content changes between consecutive frames: high values indicate pronounced frame-to-frame changes, low values spectrally stable segments. In the projection, lower spectral flux concentrates in the left region, predominantly *Sad* and *Neutral*, whereas higher spectral flux concentrates on the right, mainly *Angry*, *Surprise*, and *Happy*. Emotions with higher articulatory and energetic variability thus spread along this direction.

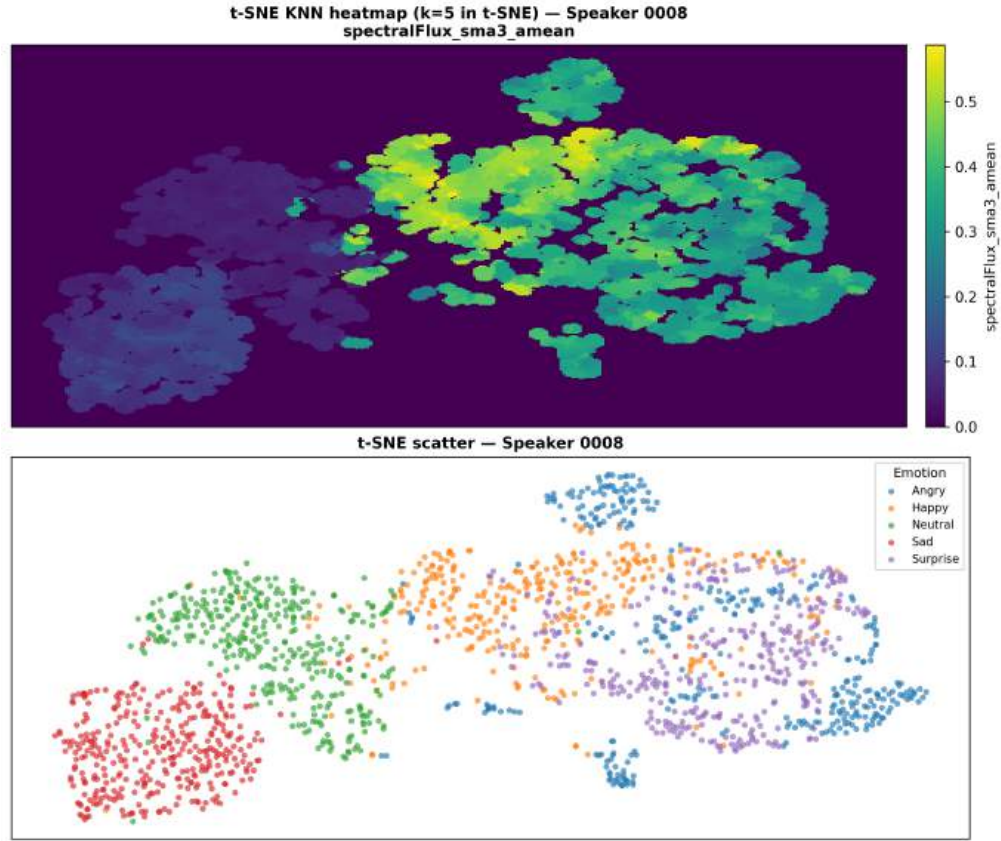


Figure 24: t-SNE projection of Speaker 0008 with heatmap of *spectralFlux_sma3_amean*, obtained by k -NN interpolation ($k = 5$) of the feature value over the 2D grid.

Loudness falling slope describes how quickly perceived loudness decreases over time, capturing how fast vocal energy dissipates after energetic peaks. The left region, dominated by *Sad* and *Neutral*, shows small falling-slope values, consistent with reduced respiratory modulation and slowly decaying loudness. The right region shows markedly higher values and contains mainly *Happy*, *Surprise*, and *Angry*, consistent with more dynamic vocal energy in which loudness rises sharply and decays rapidly under stronger respiratory impulses.

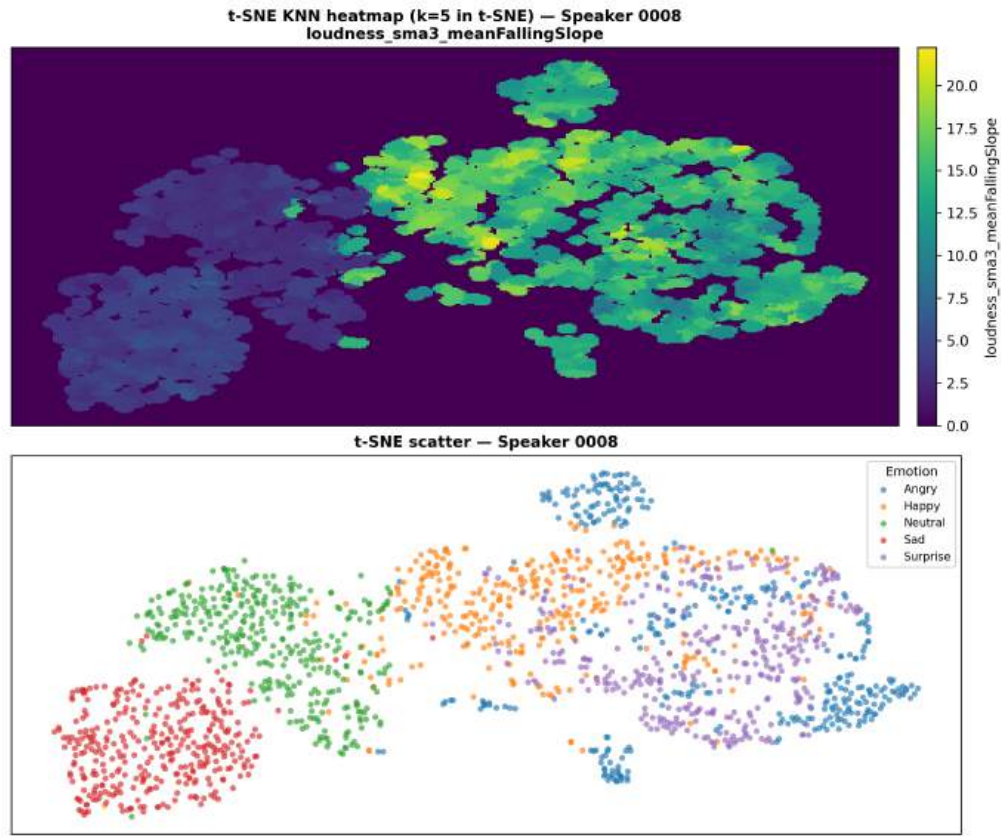


Figure 25: t-SNE projection of Speaker 0008 with heatmap of *loudness_sma3_meanFallingSlope*.

Alpha ratio quantifies the dominance of low-frequency energy over higher-frequency components, distinguishing low-frequency-dominated speech from a brighter spectral balance. The heatmap shows the lowest values in the *Neutral* and *Angry* regions, where energy concentrates at lower frequencies: for *Neutral*, consistent with controlled vocal production and limited articulatory exaggeration; for *Angry*, with strong vocal-fold tension that concentrates energy in the lower harmonics while suppressing higher frequencies. *Happy*, *Surprise*, and *Sad* tend to occupy brighter-balance regions.

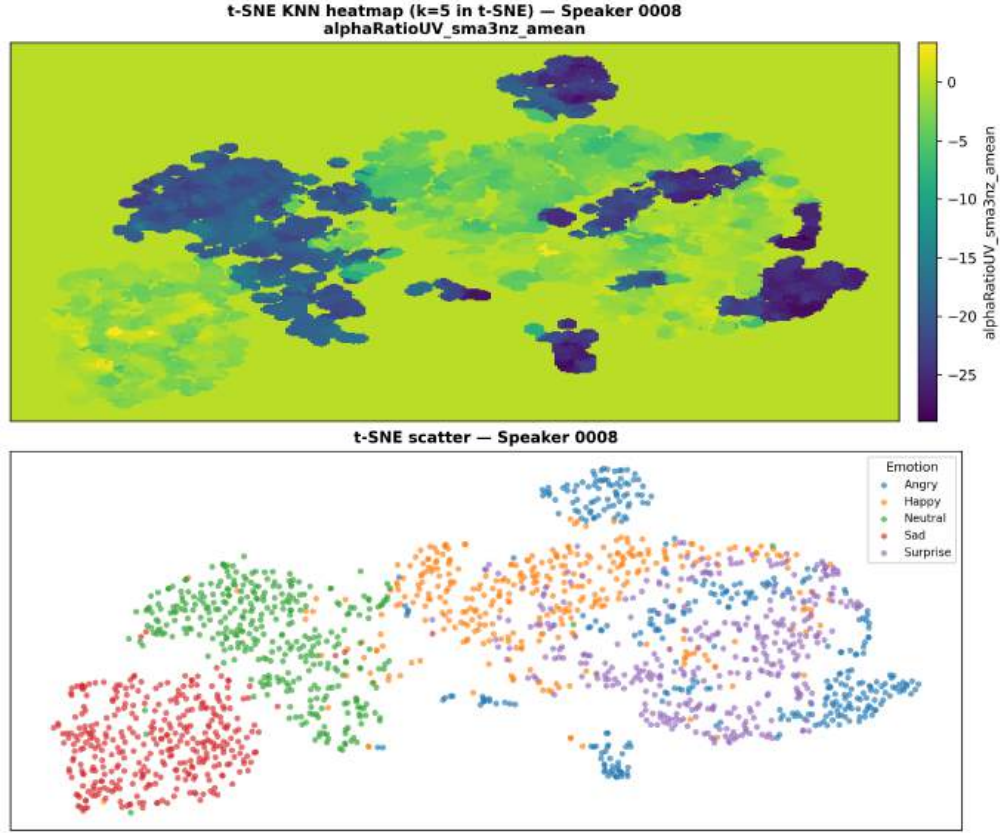


Figure 26: t-SNE projection of Speaker 0008 with heatmap of *alphaRatioUV_sma3nz_amean*.

Taken together, the three overlays suggest that the local arrangement in the t-SNE projection of Speaker 0008 aligns with interpretable acoustic dimensions of energy dynamics and spectral shape: *spectral flux* and *loudness falling slope* separate the low-energy emotions (*Sad*, *Neutral*) from the high-energy ones (*Angry*, *Surprise*, *Happy*), while the *alpha ratio* reflects the spectral balance of vocal production. These observations remain qualitative and speaker-specific, and do not yet establish whether the local acoustic structure is predictive of the emotional category; turning this intuition into a systematic, quantitative result is the subject of the future work outlined below.

6.2 Future work

The case study shows that the encoder geometry co-varies with interpretable acoustic descriptors, but only locally and for the single speaker with the clearest structure. The natural development is to make this link systematic: rather than overlaying descriptors on a 2D projection of one speaker, one would probe the encoder embeddings directly for the same eGeMAPS features

across a full corpus, testing whether the prosodic cues that appear discriminative are those the encoder uses to organize its emotion subspace.

This analysis may also help explain the residual variability in cross-corpus performance. Zero-shot accuracy varies substantially, from 94.69% on EmoDB down to 63.43% on CaFE, a gap plausibly connected to acoustic-level variability: corpora whose emotional portrayals are acoustically more pronounced may present the encoder with cleaner, better-separated prosodic patterns than those recorded under more heterogeneous conditions.

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